

Physics-augmented neural networks and projection-based reduced-order models for anisotropic hyperelasticity

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Abstract

Approximating the effective anisotropic hyperelastic response of a representative volume element (RVE) requires constitutive expressiveness, mechanical consistency, and computational efficiency. Non-intrusive neural surrogates provide direct constitutive evaluations, whereas intrusive projection-based reduced-order models (PROMs) approximate the microscopic state and equilibrium equations. This work develops a polyconvex physics-augmented neural network (PANN) construction based on paired directional features with trainable orientations, stretch powers, and determinant exponents. The features are convex in independent deformation minors and are composed with convex, componentwise nondecreasing cores realized as an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). Their isotropic reference derivatives enable affine-determinant stress normalization without compromising polyconvexity. The complete energies satisfy objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and prescribed growth conditions by construction; global nonnegative energy is assessed through a separate sufficient bound for the trained parameters. Projection-based surrogates with hyperreduction provide a complementary benchmark for structural accuracy and online cost. On a periodic cell with a rotated elliptical void, the selected polyconvex models attain approximately 0.65% held-out stress error. A tensile-coupon application illustrates structural deployment against a full-order FE² reference. The ICNN and ICKAN give displacement errors below 0.43%; the iterative nonlinear reduced model gives 0.052% error and its implemented workflow is 2.23 times faster than the optimized affine reduced model under the reported configurations. These results demonstrate the practical use of the constrained constitutive representation and quantify its accuracy–cost trade-off against reduced microscopic solvers.

Keywords: computational homogenization; hyperelasticity; physics-augmented neural networks; polyconvexity; projection-based reduced-order models; empirical cubature.

1 Introduction

The effective response of a heterogeneous material emerges from the deformation of its constituents and their interactions with the microstructure. At finite strains, this interaction can produce anisotropy and coupling between loading modes that are difficult to capture with a predetermined macroscopic constitutive law. Computational homogenization addresses this difficulty by resolving a representative volume element (RVE): an imposed deformation drives a microscopic boundary-value problem, from which effective stresses and constitutive derivatives are obtained [1]. Rather than prescribing the effective law in advance, the RVE supplies it through the solution of the underlying mechanical problem.

This flexibility comes at a computational cost. Characterizing the material over a range of deformations already requires many nonlinear finite element solves; using the same RVE within a structural calculation multiplies those solves across macroscopic integration points and load increments. Much of this repeated work concerns closely related microscopic states. Surrogate modeling seeks to exploit that relationship by computing representative solutions offline and reusing their information in less expensive online evaluations. The objective is therefore not only to reduce execution time and the resources required to store and process microscopic fields, but to do so without losing the constitutive behavior that made the RVE description useful.

There are two complementary ways to pursue this objective. Non-intrusive constitutive surrogates learn the effective response from RVE input–output data and dispense with the microscopic solve online. Intrusive projection-based reduced-order models (PROMs) instead compress the microscopic displacement field and retain projected equilibrium equations, with hyperreduction limiting the local contributions needed to evaluate them. The former offers a direct constitutive evaluation; the latter retains access to an approximate microscopic state and the given constituent law. Mechanical knowledge and learning can enter either route: physics-augmented neural networks (PANNs) embed constitutive restrictions in the learned response, while artificial neural networks (ANNs) can enrich a PROM’s displacement representation.

This work develops a polyconvex PANN for the effective anisotropic hyperelastic response of an RVE. The aim is to make the learned energy sufficiently expressive while preserving its mechanical structure, including an exactly stress-free reference state. PROM-based surrogates of the same RVE provide complementary accuracy–cost comparisons, and a tensile structural example examines how the constitutive approximations behave when deployed beyond isolated material evaluations. The constitutive construction is the central contribution; structural deployment supplies a further test of its usefulness.

1.1 Physics-augmented neural networks

The attraction of direct constitutive learning is that the expensive microscopic solver is needed to generate data, but not to evaluate the trained material model. Neural homogenization has demonstrated this separation [2], making the systematic generation of representative microscopic responses an important part of the offline workflow [3]. Once this separation is made, however, the inputs and architecture of the surrogate must carry the information that would otherwise be supplied by the microscopic problem.

For materials with memory, this information extends beyond the current deformation. Loading history can be retained through learned history-to-response maps [4] or through internal descriptors, such as the maximum directional stretches used in polymer models [5]. The data-generation burden also grows when several interacting scales must be represented; staggered training of successive two-scale surrogates provides one way to manage that burden [6]. Here we consider history-independent hyperelasticity. Removing the need to represent memory simplifies the state description, but introduces a precise constitutive requirement: the stress response must be compatible with a stored-energy function.

Learning that function, rather than fitting each stress component independently, makes the stress work conservative and gives a major-symmetric constitutive tangent by differentiation. This is a central motivation for the mechanics-informed energy formulation of As’ad et al. [7]. Expressing the energy through objective deformation measures then removes dependence on a superposed rigid rotation. In this way, PANNs shift part of the burden from learning mechanical identities from data to satisfying them through the representation itself [8, 9]. Prediction accuracy remains to be established, but potential structure and objectivity need not depend on how densely the training domain has been sampled.

Preserving a potential does not, by itself, control its curvature. At finite strains, that control must also be imposed in appropriate variables. Convexity in Green–Lagrange strain, as used in Ref. [7], is attractive because stress and tangent follow directly from the learned energy. Its implication for curvature with respect to the deformation gradient nevertheless contains a stress-dependent term [10], so it is not a global polyconvexity guarantee. Conversely, enforcing a positive-semidefinite matrix through a lower-triangular factorization [11] controls a matrix property, but does not by itself supply the integrability needed to recover a scalar energy. The modeling task is to preserve both the potential and the desired finite-strain curvature condition within one construction.

Polyconvexity provides a route to that combination. It represents the energy as a convex function of deformation minors, yielding a sufficient condition for rank-one convexity [12]. For the two-dimensional model considered here, the independent arguments are the deformation gradient and its determinant. This framework also underpins existence results when the additional coercivity and admissibility hypotheses are met [13]. We use it as a sufficient constitutive restriction, not as a necessary property of every homogenized material or a guarantee of structural uniqueness. The remaining challenge is representational: the restrictions that provide a guarantee must still leave enough freedom to describe the anisotropic RVE response.

1.2 Projection-based reduced-order models

An alternative to learning the effective energy is to retain the microscopic problem and reduce the space in which it is solved. In a conventional PROM, solution snapshots supply a reduced-order basis, usually through proper orthogonal decomposition (POD), and projected equilibrium determines the coefficients of the displacement field in that basis. This allows information from previous RVE solves to be reused without replacing the constituent law, as demonstrated in reduced finite-strain homogenization [14, 15]. Retaining a field representation also supports multiscale settings in which reduced periodic substructures must exchange

kinematic information [16]. For the present RVE, the question is how much of this field information can be retained at a small online cost.

The first limitation is the approximation space. A family of solutions may depend on only a few loading parameters and still be poorly represented by a small global linear basis. The Kolmogorov n -width makes this distinction precise: if the best worst-case error over all n -dimensional linear spaces decays slowly, adding modes remains expensive even when the solution family has a low-dimensional parametrization [17, 18]. Local bases address this difficulty by distributing the approximation among regions of the solution set [19]. Nonlinear manifolds pursue the same objective without restricting each representation to an affine space, using, for example, autoencoder mappings [20] or quadratic corrections [21].

This motivates retaining a rich POD representation while reducing the number of coefficients that must be determined independently. The PROM-ANN formulation [17] achieves this by solving for a small set of primary coordinates and using an ANN to reconstruct the secondary ones.¹ The displacement field can thus retain the contribution of many POD modes without introducing an equilibrium unknown for each one. We adopt this primary/secondary representation with an ANN closure for the RVE, seeking to retain the accuracy of the richer displacement space at the cost of a smaller nonlinear system.

Reducing the unknowns does not eliminate a second cost: evaluating the microscopic contributions entering the projected equations. Even a very small reduced system can remain expensive if its residual and tangent require constitutive evaluations throughout the original mesh. Hardware acceleration can lower the cost of those evaluations, as in GPU-assisted reduced homogenization [22]; hyperreduction instead reduces how many contributions must be evaluated. It is this second approximation that turns a PROM into a hyperreduced projection-based reduced-order model (HPROM).

Two broad families approach this task from different directions [23, 24]. Approximate-then-project methods, notably discrete empirical interpolation (DEIM) [25], reconstruct a high-dimensional nonlinear quantity from selected samples before projecting its approximation. For a mechanical model, however, accurate reconstruction of that quantity does not automatically preserve the variational structure of the reduced equations. Project-then-approximate methods target the reduced quantities directly, often by replacing full assembly with a weighted sum of selected local contributions. Optimized cubature [26], energy-conserving sampling and weighting (ECSW) [27, 28], and the empirical cubature method (ECM) [29] follow this route. The structure-preservation results for ECSW in structural dynamics [28] illustrate the value of designing the approximation around the mechanical equations, although they do not constitute a stability guarantee for every sampling-and-weighting method. For the RVE considered here, this route also has a direct computational advantage: selected finite element contributions can be evaluated on a reduced mesh using the existing constituent law. We adopt ECM within this setting.

The original ECM construction [29] obtains a sparse integration rule by selecting spatial points and nonnegative weights that reproduce a training space of integrands. Its economy comes from replacing the many contributions in the full discretization by a much smaller weighted subset. The resulting rule nevertheless retains two restrictions: points are selected from a discrete candidate set, and one weight vector must serve the entire training space. Subsequent developments show that either restriction can be relaxed to seek greater compression. The continuous empirical cubature method (CECM) [30] takes an initially feasible discrete rule and removes points while readjusting both the positions and weights of those that remain. Here the extra freedom is spatial: the integration points are no longer confined to the original candidates.

For reduced models built from local spaces or nonlinear manifolds, the second restriction becomes especially relevant. Once different regions of the solution set have their own local bases, requiring one fixed weight vector to reproduce the conditions from all of them can impose an unnecessarily global integration problem. Subspace-adaptive weights ECM (SAW-ECM) [31] exploits this observation by retaining a common set of points while assigning different nonnegative weights to each local basis. As the local representation was refined in the examples of that study, the required common support decreased. The gain came from adapting how the selected points contribute, rather than from moving their locations.

Manifold-adaptive weights ECM (MAW-ECM) [32] carries this idea from a collection of local spaces to a continuously parameterized solution manifold: latent coordinates replace the local-basis index, and each retained spatial entity receives a weight field over the latent domain. The integration rule can thus follow the reduced state instead of reproducing the global linear span of all training conditions with one fixed vector. This is the connection to the nonlinear closure above. The MAW-ECM study illustrates it with an RBF-based reconstruction; the present work uses an ANN closure and neural weight predictors. The adaptive-weight principle concerns the manifold and its integrands, not a particular regression architecture.

These developments motivate two complementary reduced evaluators for the present comparison: an affine HPROM with fixed ECM, and a nonlinear-manifold HPROM-ANN with adaptive weights. A direct mode,

¹The closure need not be neural: the latent-space framework of Ares De Parga et al. [18] also considers Gaussian-process regression (GPR) and radial-basis-function (RBF) closures.

D-HPROM-ANN, additionally uses strain-informed coordinates to reconstruct the microscopic response without reduced equilibrium iterations. Comparing these modes examines how displacement compression, integration compression, and online equilibrium work affect accuracy and cost. Their constitutive outputs must still be assessed on their own terms: retaining the constituent law and positive cubature weights does not alone establish that every approximate effective stress derives from one energy.

1.3 Feature representation and position relative to prior work

The two surrogate routes therefore place the approximation in different parts of the problem. A PROM compresses the microscopic state while continuing to evaluate the given material law; an energy-based PANN must represent the effective law itself. For an anisotropic response, this requires more than a flexible scalar network. The inputs must express directional dependence while remaining compatible with objectivity and the desired convexity condition.

Structural tensors provide a way to encode material directions in objective scalar quantities and have enabled anisotropic polyconvex energies [33]. The directional description matters as much as the scalar response assigned to it: models of dispersed fibers, for example, account for a distribution of orientations rather than one preferred axis [34]. In a learned constitutive law, the corresponding feature map determines which distinctions between deformation states the network can use. Encoding directions is therefore part of the model, not merely a preprocessing step.

An ICNN provides a convex scalar core [35], but composing that core with nonlinear features requires additional care. A sufficient construction uses features convex in the independent deformation minors and a core that is also nondecreasing in each feature [8]. The latter condition links admissibility to expressiveness. If two states satisfy $\mathbf{z}(\mathbf{F}_a) \leq \mathbf{z}(\mathbf{F}_b)$ componentwise, a monotone core must satisfy $H(\mathbf{z}(\mathbf{F}_a)) \leq H(\mathbf{z}(\mathbf{F}_b))$. Increasing its width cannot reverse that ordering. When reference corrections and growth terms are present, the comparison concerns the part of the energy represented by the core, but the restriction remains. An unsuitable feature family can therefore prevent an otherwise admissible architecture from fitting the target response.

This is not merely a formal possibility. In the cubic-lattice example of Klein et al. [8], an invariant-based polyconvex model has difficulty representing shear, while a deformation-gradient-based formulation improves the fit. The latter uses data augmentation for approximate objectivity, whereas the invariant model is exactly objective and performs well for their transversely isotropic example. These results point to the representation, rather than polyconvexity alone, as a potential source of limited accuracy. They motivate enlarging the admissible feature family before concluding that a convex architecture is inadequate for a material.

Allowing the directional description to adapt during training is one way to gain that flexibility. Generalized-structure-tensor models learn aspects of the anisotropy [36], and objective triclinic polyconvex constructions show that a nontrivial material symmetry group need not be prescribed in advance [37]. Yet directional flexibility must also be reconciled with the reference state. A learned anisotropic energy can produce a nonzero reference stress, and cancelling that stress with an arbitrary strain correction can destroy polyconvexity. The compatible normalization and growth terms developed in Ref. [9] show why these corrections belong to the constitutive construction itself. This leads to the design question pursued here: can the directional features be made flexible while controlling their reference derivatives so that stress normalization remains admissible?

Once the features convey the required directional information, the scalar core must resolve how the energy varies with them. This flexibility can be placed in the coefficients of admissible energy terms, as in constitutive artificial neural networks (CANNs) [38], or in learned monotone energy derivatives [39]. For the present construction, an ICNN supplies a compatible core through weighted compositions of fixed convex activations. A spline-based Kolmogorov–Arnold network instead learns the shapes of its univariate edge functions, providing a complementary way to distribute that flexibility. Constitutive KAN formulations [40] motivate this alternative, but preserving the feature-composition proof requires more than using a scalar energy output: each edge must also respect the required convexity and monotonicity. Monotonic input-convex KANs (ICKANs) provide this connection [41]. We therefore assess the paired features with both cores and use analytically joined splines so that the admissibility argument also covers their tails. The comparison tests the scalar approximation while retaining the same type of mechanically constrained feature representation.

1.4 Present contribution

We address the feature and normalization requirements together through paired directional features with trainable orientations, stretch powers, and determinant exponents constrained by explicit joint-convexity bounds. Pairing makes each feature’s strain derivative at the undeformed state proportional to the identity tensor, even though its response away from that state remains anisotropic. A single scalar correction affine in the determinant can then cancel the complete reference stress without tying that correction to the

independently adjustable volumetric stiffness. The feature family thus gains flexibility while retaining an analytically controlled reference state.

The family is coupled to a smooth ICNN and an integrated-hat spline ICKAN, both convex and componentwise nondecreasing in the features. For either core, the proof covers the complete energy, including normalization and growth terms, and establishes objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and the specified growth limits. The spline realization uses analytically matched tails. Global nonnegative energy is addressed separately by a sufficient lower bound evaluated for the trained parameters. The contribution builds on established convex-network and affine-determinant normalization principles; it is their realization with the paired anisotropic family and its accompanying analytical conditions.

Figure 1 summarizes the resulting constitutive construction. RVE responses train a family of paired directional features and either of two monotone convex cores. The learned contribution is then completed by analytic reference and growth terms, producing a stored energy from which stress and tangent follow by differentiation. The numerical study places this construction beside a flexible energy reference to interpret the cost of the guarantees, and beside reduced microscopic evaluators to contextualize structural accuracy and online cost.

1.5 Assessment strategy and scope

The examples address two successive questions. First, how accurately can the constrained energy represent anisotropic RVE responses, and what does learning the paired features contribute? The constitutive assessment examines independent response errors and the fixed/learned-feature comparison. Derivative checks and saved-parameter energy bounds complement the analytical construction, following the broader motivation to examine constitutively relevant derivatives as well as fitting errors [42]. Sampled curvature tests check the implementation; the polyconvexity guarantee comes from the proof.

Second, how do those constitutive approximations behave within a nonlinear structural calculation? A tensile coupon supplies this deployment test. In the first-order two-level finite element formulation, denoted FE^2 , each macroscopic integration point queries the same RVE response. Replacing that query by each surrogate propagates its approximation through a common equilibrium problem. A nested full-order calculation (FOM-FE^2) supplies the reference, and field-level comparisons complement aggregate displacement and stress errors [43]. The example thus connects material-level assessment to structural accuracy and online cost without making the development of a new FE^2 scheme the subject of the paper.

The mechanical guarantees and the numerical comparisons answer distinct questions. The construction establishes the properties of the energy, while independent RVE and structural evaluations assess its approximation accuracy and computational usefulness. Additional closed-cycle and curvature diagnostics are documented in the supplementary material.

The structural comparison concerns material A under plane-strain kinematics and one loading path. FOM-FE^2 agreement measures surrogate error within the adopted homogenization model, not its accuracy against a structure resolving every microscopic cell. The study assesses the feature family, normalization, and core together; isolating their individual contributions requires matched-budget ablations. Likewise, the reported online times compare the implemented workflows at the stated process/thread counts and repetition levels, excluding offline data generation and training. They do not quantify memory savings or an offline-inclusive break-even point. Within this scope, the aim is to establish the usefulness and constitutive properties of the proposed representation, not universality across microstructures, general three-dimensional completeness, or guaranteed robustness under microscopic instability.

Section 2 defines the homogenization problem and work-conjugate variables. Section 3 develops the PROM-based surrogates. Section 4 introduces the admissibility criteria, constructs the learned energies, and proves their properties before specifying the implementation checks. Sections 5 and 6 report the constitutive and structural assessments, respectively; Section 7 closes the paper.

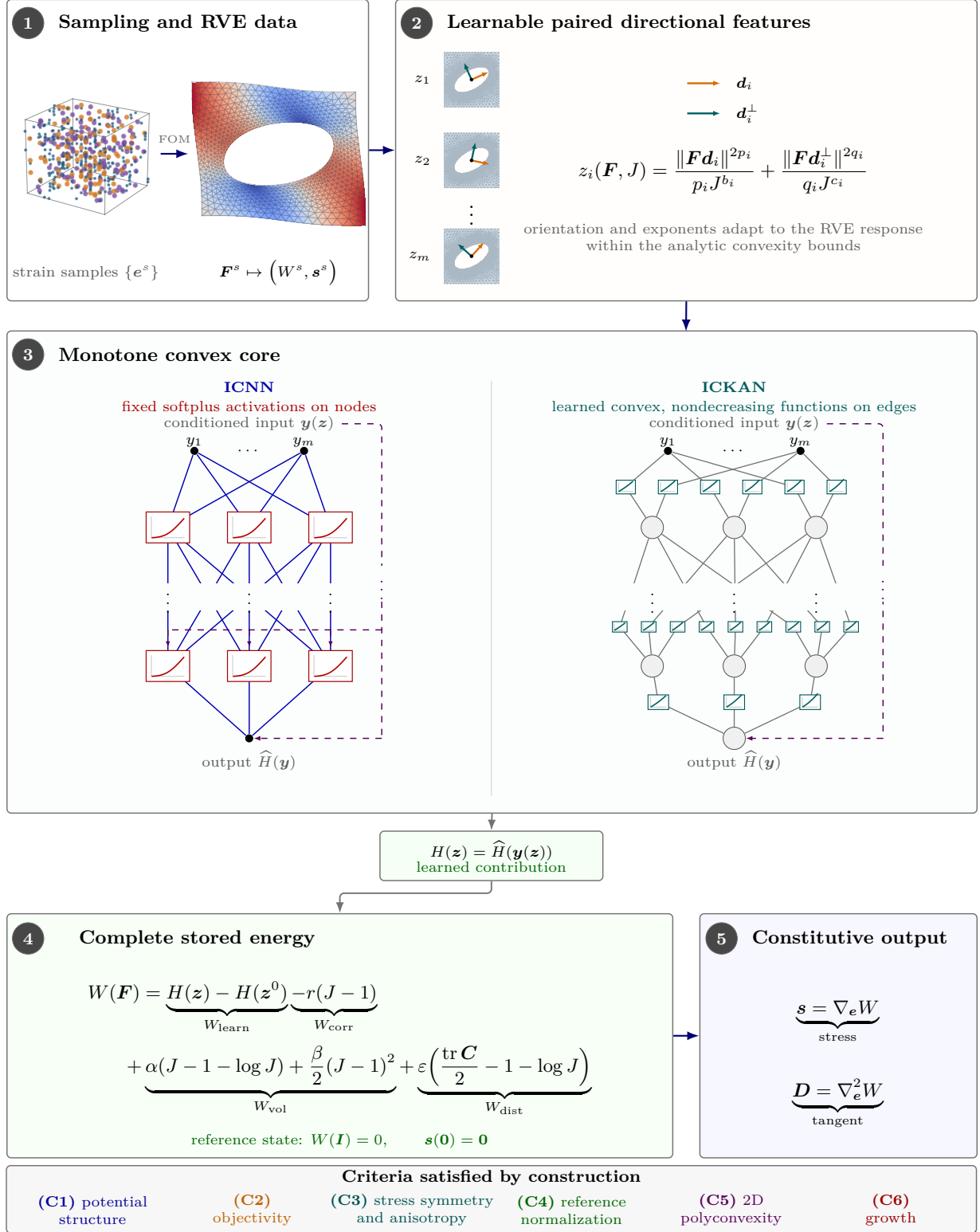


Figure 1: Overview of the proposed constitutive construction. Sampled strain states are evaluated with the periodic RVE model to supply energy and stress data. These responses train learnable paired directional features and a monotone convex ICNN or ICKAN core. Analytic reference and growth terms complete the stored energy; differentiation then supplies the work-conjugate stress and material tangent. The stated properties follow from the complete construction rather than from sampled numerical checks.

2 Finite-strain homogenization and the reference problem

The surrogates considered here approximate the same constitutive problem: given a macroscopic deformation, determine the effective response of a periodic RVE. The full-order model obtains this response by solving microscopic equilibrium. We first define its input, boundary conditions, and effective outputs, and then identify the stress–tangent interface needed for structural deployment. The cell geometry and material parameters are specified in the numerical examples.

2.1 Kinematics and work-conjugate variables

Let $\bar{\mathbf{F}} \in \text{GL}^+(2)$ denote the prescribed in-plane macroscopic deformation gradient. Its associated right Cauchy–Green tensor and Green–Lagrange strain are

$$\mathbf{C} = \bar{\mathbf{F}}^T \bar{\mathbf{F}}, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad J = \det \bar{\mathbf{F}} > 0. \quad (1)$$

Here $\mathbf{C}$ and $\mathbf{E}$ describe the macroscopic state; the subscript μ will distinguish microscopic quantities. We adopt plane strain: the corresponding three-dimensional gradient is $\text{diag}(\bar{\mathbf{F}}, 1)$, and the same embedding applies locally to $\mathbf{F}_\mu$. This constrains the out-of-plane deformation, not the out-of-plane stress.

The constitutive input is the three-component engineering strain vector. With $\bar{\mathbf{S}}$ the effective second Piola stress, the work-conjugate convention is

$$\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (\bar{S}_{11}, \bar{S}_{22}, \bar{S}_{12})^T, \quad \bar{\mathbf{S}} : d\mathbf{E} = \mathbf{s} \cdot d\mathbf{e}. \quad (2)$$

Thus $\gamma_{12} = 2E_{12}$ is paired with $\bar{S}_{12}$, with no additional factor of two in the stress vector. For a differentiable effective energy density $\bar{W}(\mathbf{e})$, this convention gives $\mathbf{s} = \partial_{\mathbf{e}} \bar{W}$ and $\mathbf{D} = \partial_{\mathbf{e}} \mathbf{s}$. It is used consistently in training, work integrals, and tangent checks.

An admissible strain input must satisfy $\mathbf{I} + 2\mathbf{E} \succ 0$. We query the RVE with the representative

$$\bar{\mathbf{F}}(\mathbf{e}) = \sqrt{\mathbf{I} + 2\mathbf{E}}, \quad (3)$$

where the square root is symmetric positive definite. Objectivity allows a superposed macroscopic rotation to be removed without changing the energy or second Piola stress. This choice does not assume isotropy: material directions remain fixed in the reference configuration. Nor does it require the local microscopic gradients to be symmetric.

2.2 Periodic microscopic equilibrium

Let Y denote the reference periodic cell and $Y_s \subset Y$ its solid part. The prescribed $\bar{\mathbf{F}}$ determines the affine part of the microscopic deformation; the fluctuation $\mathbf{w}$ is unknown:

$$\boldsymbol{\varphi}(\mathbf{X}) = \bar{\mathbf{F}}\mathbf{X} + \mathbf{w}(\mathbf{X}), \quad \mathbf{F}_\mu = \bar{\mathbf{F}} + \nabla \mathbf{w}, \quad (4)$$

with gradients taken in the reference configuration. Periodicity requires $\mathbf{w}(\mathbf{X}^+) = \mathbf{w}(\mathbf{X}^-)$ at paired points on opposite cell faces. Consequently, the microscopic displacement $\mathbf{u}_\mu = \boldsymbol{\varphi} - \mathbf{X}$ satisfies

$$\mathbf{u}_\mu(\mathbf{X}^+) - \mathbf{u}_\mu(\mathbf{X}^-) = (\bar{\mathbf{F}} - \mathbf{I})(\mathbf{X}^+ - \mathbf{X}^-). \quad (5)$$

The finite-strain displacement jump therefore depends on $\bar{\mathbf{F}} - \mathbf{I}$, not directly on $\mathbf{E}$. Setting one fluctuation value to zero removes the arbitrary translation; it does not clamp the entire cell boundary. The internal void surface is traction-free.

Let $\psi_\mu(\mathbf{X}, \mathbf{F}_\mu)$ be the solid’s objective hyperelastic energy per unit reference volume and $\mathbf{P}_\mu = \partial_{\mathbf{F}_\mu} \psi_\mu$ its first Piola stress. In the absence of microscopic body forces, equilibrium requires

$$\int_{Y_s} \mathbf{P}_\mu : \nabla \delta \mathbf{w} \, dV = 0 \quad (6)$$

for every admissible periodic test fluctuation. Periodic kinematics and the associated traction balance provide the standard work-consistent cell problem [1]. After finite element discretization and elimination of the periodic constraints, let $\mathbf{a}$ collect the independent fluctuation coordinates. The full-order equilibrium problem is

$$\mathbf{r}(\mathbf{a}, \mathbf{e}) = \partial_{\mathbf{a}} \Pi_\mu(\mathbf{a}, \mathbf{e}) = \mathbf{0}, \quad \Pi_\mu = \int_{Y_s} \psi_\mu(\mathbf{X}, \mathbf{F}_\mu) \, dV, \quad (7)$$

where the strain dependence enters through Eq. (3). Solving Eq. (7) gives $\mathbf{a}^*(\mathbf{e})$ and hence the microscopic fields. The microscopic deformation must also remain orientation-preserving, $\det \mathbf{F}_\mu > 0$.

2.3 Effective energy and stress

On the selected equilibrium branch, the effective energy and stresses are

$$\bar{W}(\mathbf{e}) = \frac{\Pi_\mu(\mathbf{a}^*(\mathbf{e}), \mathbf{e})}{|Y|}, \quad \bar{\mathbf{P}} = \frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu \, dV, \quad \bar{\mathbf{S}} = \bar{\mathbf{F}}^{-1} \bar{\mathbf{P}}. \quad (8)$$

The normalizing volume $|Y|$ includes the void: these are properties of the porous cell, not of its solid fraction alone. For the two-dimensional discretization, the same reference thickness enters the integration weights and the cell volume, and cancels in their ratio. The stress conversion in Eq. (8) is performed *after* averaging the first Piola stress. In general, averaging microscopic second Piola stresses does not produce $\bar{\mathbf{S}}$, because $\mathbf{F}_\mu$ varies within the cell.

The connection between energy and averaged stress follows from equilibrium. For a macroscopic variation, $\delta \mathbf{F}_\mu = \delta \bar{\mathbf{F}} + \nabla \delta \mathbf{w}$; Eq. (6) removes the fluctuation contribution and yields

$$\frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu : \delta \mathbf{F}_\mu \, dV = \bar{\mathbf{P}} : \delta \bar{\mathbf{F}} = \bar{\mathbf{S}} : \delta \mathbf{E} = \mathbf{s} : \delta \mathbf{e}. \quad (9)$$

This is the Hill–Mandel work-consistency condition [1]; it also establishes $\mathbf{s} = \partial_{\mathbf{e}} \bar{W}$ on a smooth branch. The subsequent constitutive constructions omit overbars when the effective, rather than microscopic, response is unambiguous.

The Newton solver follows an equilibrium branch, possibly through load continuation. Convergence establishes stationarity, not global minimization of Π_μ or uniqueness of the cell response. We use the resulting smooth branch as the reference constitutive response over the investigated states; we do not assume that microscopic polyconvexity guarantees a polyconvex homogenized energy.

2.4 Consistent constitutive tangent

The tangent must account for the change in microscopic equilibrium as the input strain changes. To make this dependence explicit, write the stress evaluator before substitution of the equilibrium state as $\hat{\mathbf{s}}(\mathbf{a}, \mathbf{e})$, so that $\mathbf{s}(\mathbf{e}) = \hat{\mathbf{s}}(\mathbf{a}^*(\mathbf{e}), \mathbf{e})$. If $\mathbf{r}_{,\mathbf{a}}$ is nonsingular on the selected branch, implicit differentiation of Eq. (7) gives

$$\frac{d\mathbf{a}^*}{d\mathbf{e}} = -\mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}, \quad \mathbf{D} = \hat{\mathbf{s}}_{,\mathbf{e}} - \hat{\mathbf{s}}_{,\mathbf{a}} \mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}. \quad (10)$$

All partial derivatives are evaluated at equilibrium, holding the other argument fixed. The first term differentiates the stress at fixed fluctuation coordinates; the second accounts for microscopic relaxation. Omitting that correction generally does not give the derivative of the deployed constitutive response. For the smooth hyperelastic branch, the result is also $\partial_{\mathbf{e}}^2 \bar{W}$.

No explicit matrix inverse is needed in Eq. (10): the sensitivities follow from linear systems with the converged microscopic tangent. The reference implementation uses one nonlinear equilibrium solution followed by three sensitivity right-hand sides, one per strain component. Independent finite-difference checks perturb the input and re-solve equilibrium. Such checks assess the constitutive linearization, not the physical accuracy of the homogenization assumptions. The same consistency requirement applies to each surrogate: its tangent must differentiate its own stress evaluation.

2.5 Macroscopic equilibrium and the constitutive interface

Structural deployment provides an application of this constitutive map. Let $\mathbf{U}$ collect the free macroscopic nodal displacements, and let $\mathbf{e}_g(\mathbf{U})$ be the engineering Green strain at integration point g . With $\mathbf{B}_g = \partial \mathbf{e}_g / \partial \mathbf{U}$, reference-volume integration weights Ω_g , and dead external loads $\mathbf{f}_{\text{ext}}$, equilibrium reads

$$\mathbf{R}_M(\mathbf{U}) = \sum_g \Omega_g \mathbf{B}_g^T \mathbf{s}_g(\mathbf{e}_g(\mathbf{U})) - \mathbf{f}_{\text{ext}} = \mathbf{0}. \quad (11)$$

Its tangent contains material and geometric contributions,

$$\mathbf{K}_M = \sum_g \Omega_g \left[\mathbf{B}_g^T \mathbf{D}_g \mathbf{B}_g + \sum_{\alpha=1}^3 (s_g)_\alpha \frac{\partial^2 (e_g)_\alpha}{\partial \mathbf{U} \partial \mathbf{U}} \right]. \quad (12)$$

The first term uses the constitutive tangent; the second arises because the Green strain is nonlinear in the displacements. A structural Newton solver therefore needs both $\mathbf{s}_g$ and $\mathbf{D}_g$, irrespective of how the constitutive response is computed.

In FOM–FE², each constitutive call invokes the full periodic RVE problem. The iterative HPROM variants replace it with projected equilibrium on reduced coordinates and hyperreduced integration. The direct decoder and learned constitutive models bypass microscopic equilibrium iterations. These alternatives are developed in the following sections; all supply the same stress–tangent interface while the macroscopic mesh, loading, and equilibrium algorithm remain unchanged.

The nested FOM–FE² solution is the structural reference for measuring surrogate errors at fixed macro and micro discretizations. It is not a direct numerical simulation resolving every pore in the coupon. Homogenization error, boundary-layer effects, and mesh-discretization error are therefore distinct from the surrogate errors reported below. The cell itself is an ideal periodic material model, without a claim of statistical representativeness for a random microstructure. The suffix FE² in archived model names identifies the common structural test; a direct neural constitutive evaluation is not a nested microscopic finite element solve.

3 Projection-based reduced-order models

The intrusive route reduces the cost of the RVE problem while retaining evaluations of the microscopic constitutive law. Two approximations play different roles. A reduced displacement representation decreases the number of equilibrium unknowns; hyperreduction decreases the number of element contributions evaluated to assemble the reduced equations and effective outputs. We first construct affine and nonlinear-manifold PROMs, then introduce fixed and manifold-adaptive cubature. Finally, we distinguish solving reduced equilibrium from evaluating the same decoder directly.

3.1 The constrained high-dimensional model

We follow the nomenclature of Ref. [18]: the high-dimensional model (HDM) has state dimension N ; a PROM solves for generalized coordinates in a reduced approximation space; an HPROM additionally hyperreduces the evaluation of its projected operators. HDM and FOM denote the same unreduced microscopic discretization here; FOM is retained in the archived name FOM–FE².

It is useful to distinguish the physical nodal displacement vector $\mathbf{d} \in \mathbb{R}^{N_{\text{dof}}}$ from the independent state $\mathbf{u} \in \mathbb{R}^N$ obtained after enforcing periodicity and removing rigid translation. In the deployed convention,

$$\mathbf{d}(\mathbf{u}, \mathbf{e}) = \mathbf{T}\mathbf{u} + \mathbf{g}_{\text{per}}(\mathbf{e}), \quad \mathbf{R}(\mathbf{u}; \mathbf{e}) = \mathbf{T}^T \mathbf{f}_{\text{int}}(\mathbf{d}(\mathbf{u}, \mathbf{e})) = \mathbf{0}. \quad (13)$$

Here $\mathbf{T}$ identifies periodic degrees of freedom, and the lifting $\mathbf{g}_{\text{per}}$ supplies the strain-dependent displacement jumps and anchor values defined in Section 2. Thus any independent state $\mathbf{u}$ produces a physical field satisfying the prescribed kinematic constraints. If $\mathbf{K}_{\text{phys}} = \partial_{\mathbf{d}} \mathbf{f}_{\text{int}}$ is the assembled physical stiffness, the constrained HDM tangent is $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R} = \mathbf{T}^T \mathbf{K}_{\text{phys}} \mathbf{T}$. We use the internal-force sign convention for all residuals.

The snapshots used below retain the affine part of the independent displacement. They therefore differ from the fluctuation coordinates $\mathbf{a}$ of Eq. (7): if $\mathbf{u}_{\text{aff}}(\mathbf{e})$ is the affine displacement at the independent nodes, then $\mathbf{a} = \mathbf{u} - \mathbf{u}_{\text{aff}}(\mathbf{e})$. This is a change of coordinates, not a change of boundary conditions. The snapshot convention and lifting must be used together; throughout this section, $\mathbf{u}$ denotes the affine-retaining state of Eq. (13).

We retain this part to preserve load-correlated coordinates for the strain-informed identification below; preliminary modal-coordinate fits favored this convention over fluctuation-only snapshots. This is a representation choice, not a requirement of periodicity or of direct evaluation: a fluctuation decoder could instead be combined with an explicit affine lifting. Although $\mathbf{u}_{\text{aff}} = (\bar{\mathbf{F}}(\mathbf{e}) - \mathbf{I})\mathbf{X}$ is nonlinear in $\mathbf{e}$, the symmetric representative $\bar{\mathbf{F}}$ makes these affine fields linear combinations of at most three fixed nodal fields. The convention can nevertheless change the snapshot spectrum and its normalization. The retained POD dimension therefore characterizes the chosen representation and tolerance, not an intrinsic material dimension.

3.2 POD approximation and the conventional PROM

Let $\mathbf{u}^s$ be the HDM solution at a sampled macroscopic strain $\mathbf{e}^s$, with $s = 1, \dots, N_s$. Given a reference state $\mathbf{u}_{\text{ref}}$, form

$$\mathbf{S}_{\text{snap}} = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}}, \dots, \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}] = \mathbf{U}_S \boldsymbol{\Sigma}_S \mathbf{Y}_S^T. \quad (14)$$

The subscript distinguishes the snapshot matrix from the second Piola stress $\mathbf{S}$. Proper orthogonal decomposition (POD) selects the columns of $\mathbf{U}_S$ associated with the largest singular values to form a reduced-order basis (ROB). For an orthonormal n_{tra} -column basis $\mathbf{V}_{\text{tra}}$, the conventional affine approximation is

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}, \quad \mathbf{q}_{\text{tra}} \in \mathbb{R}^{n_{\text{tra}}}, \quad \mathbf{V}_{\text{tra}}^T \mathbf{V}_{\text{tra}} = \mathbf{I}. \quad (15)$$

Instead of adjusting all N displacement coordinates independently, the model adjusts $n_{\text{tra}} \ll N$ modal amplitudes. The term *linear PROM* refers to this affine approximation space; the constitutive law and reduced equilibrium equations remain nonlinear.

For a Euclidean POD, the relative squared snapshot reconstruction error is

$$\epsilon_{\text{POD}}^2(n_{\text{tra}}) = \frac{\sum_{j>n_{\text{tra}}} \sigma_j^2}{\sum_j \sigma_j^2}. \quad (16)$$

This is a compression criterion for the supplied snapshots, not an error bound for stresses at unsampled strains. Force evaluation, constitutive differentiation, and equilibrium conditioning can make small displacement components important.

At a fixed input $\mathbf{e}$, admissible reduced variations are $\delta \tilde{\mathbf{u}} = \mathbf{V}_{\text{tra}} \delta \mathbf{q}_{\text{tra}}$. Requiring the HDM residual to perform zero virtual work against these variations gives the Galerkin equations and their tangent:

$$\mathbf{r}_{\text{tra}}(\mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{V}_{\text{tra}}^T \mathbf{R}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{0}, \quad \mathbf{K}_{\text{tra}} = \mathbf{V}_{\text{tra}}^T \mathbf{K} \mathbf{V}_{\text{tra}}. \quad (17)$$

The system has n_{tra} unknowns, but assembling $\mathbf{R}$ and $\mathbf{K}$ on every microscopic element still scales with the full discretization. This is the cost removed by hyperreduction, not by POD alone.

3.3 Nonlinear reduction through latent-space closure

For a chosen displacement representation, a low-dimensional loading family may occupy a much larger linear approximation space. Latent-space closure exploits this distinction without attributing the required mode count solely to material physics: it retains a sufficiently rich displacement span, but learns dependencies among its modal amplitudes so that fewer of them remain equilibrium unknowns [17, 18]. Let $n_{\text{tot}} = n + \bar{n}$ be the dimension of that span, partitioned into mutually orthogonal primary and secondary ROB:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad \bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{n}}, \quad (18)$$

$$\mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}, \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}, \quad \bar{\mathbf{V}}^T \bar{\mathbf{V}} = \mathbf{I}. \quad (19)$$

The associated primary and secondary generalized coordinates are

$$\mathbf{q}^s = \mathbf{V}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}), \quad \bar{\mathbf{q}}^s = \bar{\mathbf{V}}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}). \quad (20)$$

The primary coordinates $\mathbf{q}$ are retained as independent unknowns. Instead of also solving for $\bar{\mathbf{q}}$, we learn the secondary-coordinate closure $\mathcal{N}: \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ from pairs $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$. This closure can be constructed using ANNs, Gaussian process regression (GPR), or radial basis function (RBF) interpolation [18]. We use an ANN to maintain a common neural parametrization across the learned components of this study, not because latent-space closure requires one. With network parameters $\boldsymbol{\eta}$, the resulting decoder and its tangent are

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q}; \boldsymbol{\eta}), \quad \mathbf{B}(\mathbf{q}) = \partial_{\mathbf{q}} \tilde{\mathbf{u}} = \mathbf{V} + \bar{\mathbf{V}} \mathcal{N}_{,\mathbf{q}}. \quad (21)$$

The correction $\bar{\mathbf{V}} \mathcal{N}(\mathbf{q}; \boldsymbol{\eta})$ turns the affine primary approximation into a nonlinear manifold. The ANN predicts neither stresses nor a macroscopic energy: it predicts the $\bar{n}$ secondary amplitudes used to reconstruct a microscopic displacement. These modes are discarded only as independent equilibrium unknowns, not from the reconstructed field. This is the PROM-ANN construction; no hyperreduction has yet been introduced.

For orthonormal blocks, the error at a snapshot separates into closure error and a component outside the total ROB:

$$\|\mathbf{u}^s - \tilde{\mathbf{u}}(\mathbf{q}^s)\|_2^2 = \|\bar{\mathbf{q}}^s - \mathcal{N}(\mathbf{q}^s)\|_2^2 + \|(\mathbf{I} - \mathbf{V}_{\text{tot}} \mathbf{V}_{\text{tot}}^T) (\mathbf{u}^s - \mathbf{u}_{\text{ref}})\|_2^2. \quad (22)$$

This explains the low-dimensional targets: the ANN approximates secondary coordinates, while the ROB performs the displacement reconstruction. It also separates limitations that cannot be repaired by the same operation. Increasing network capacity cannot recover a component excluded from $\mathbf{V}_{\text{tot}}$; adding secondary modes cannot make a non-single-valued primary-to-secondary relation single-valued.

The closure is fitted by minimizing the unscaled secondary-coordinate mean-square error,

$$\mathcal{L}_{\text{cl}}(\boldsymbol{\eta}) = \frac{1}{|\mathcal{I}_{\text{fit}}| \bar{n}} \sum_{s \in \mathcal{I}_{\text{fit}}} \|\mathcal{N}(\mathbf{q}^s; \boldsymbol{\eta}) - \bar{\mathbf{q}}^s\|_2^2. \quad (23)$$

Input standardization is absorbed into the closure map, but the secondary outputs are not standardized mode by mode. Because $\bar{\mathbf{V}}$ is orthonormal, this loss measures the independent-displacement error within the secondary space, up to its constant normalization. It is not a strain-energy norm or a stress-error

guarantee. The implemented network uses smooth tanh activations, allowing the decoder derivatives required by projection and constitutive differentiation.

For the comparison below, the affine and nonlinear models share the same retained span, with $n_{\text{tot}} = n_{\text{tra}}$. The nonlinear model represents an n -dimensional subset of this space, not every displacement the affine PROM can represent. The dimensions and reconstruction diagnostics are reported with the RVE example.

3.3.1 Strain-informed primary coordinates

A useful primary coordinate system must distinguish the displacement states whose secondary coordinates are to be predicted. The leading POD coefficients maximize captured snapshot energy, but that criterion alone does not ensure a well-conditioned primary-to-secondary relation. We use the input-informed decoder construction introduced alongside MAW-ECM in Ref. [32], with the three macroscopic strain components as inputs and $n = 3$. This construction is independent of the cubature method: it rotates the retained POD span to identify coordinates approximately aligned with those inputs.

Let $\Phi \in \mathbb{R}^{N \times n_{\text{tot}}}$ be the initial orthonormal POD basis, and collect its snapshot coefficients and the corresponding strain inputs as

$$\mathbf{Q}_{\text{POD}} = \Phi^T \mathbf{S}_{\text{snap}}, \quad \mathbf{M} = [\mathbf{e}^1, \dots, \mathbf{e}^{N_s}]. \quad (24)$$

The least-squares identification

$$\mathbf{T}_m \in \arg \min_{\mathbf{H}} \|\mathbf{H} \mathbf{Q}_{\text{POD}} - \mathbf{M}\|_F^2, \quad \mathbf{T}_m = \mathbf{M} \mathbf{Q}_{\text{POD}}^\dagger, \quad (25)$$

uses the Moore–Penrose pseudoinverse to select the minimum-norm solution. A compact SVD,

$$\Phi \mathbf{T}_m^T = \mathbf{V} \Sigma_m \mathbf{Z}_m^T, \quad \mathbf{A}_m = \Sigma_m^{-1} \mathbf{Z}_m^T, \quad (26)$$

defines the primary orthonormal block, assuming rank three. The secondary block is an orthonormal complement *within* the retained POD span, ordered by its snapshot singular values. The coordinates used in the implementation are $\xi = \mathbf{A}_m^{-1} \mathbf{q}$, with closure $\hat{\mathcal{N}}(\xi)$. Consequently, Eq. (21) becomes

$$\tilde{\mathbf{u}}(\xi) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \xi + \bar{\mathbf{V}} \hat{\mathcal{N}}(\xi), \quad \mathcal{N}(\mathbf{q}) = \hat{\mathcal{N}}(\mathbf{A}_m^{-1} \mathbf{q}). \quad (27)$$

The SVD gives $\mathbf{A}_m^{-1} = \mathbf{Z}_m \Sigma_m$, hence $\xi^s = \mathbf{T}_m \mathbf{Q}_{\text{POD}}^s$ for the projected snapshots. Their proximity to $\mathbf{e}^s$ is a fitted relation, not a kinematic identity. In particular, the input $\mathbf{e}$ prescribes the periodic lifting, while ξ parametrizes the unknown displacement. The iterative model solves for ξ at fixed $\mathbf{e}$; it does not impose their equality. In these coordinates its decoder tangent is $\mathbf{B}_\xi = \mathbf{B} \mathbf{A}_m$, so the change of variables preserves the projected equilibrium problem. Below, formulas written in $\mathbf{q}$ are evaluated with this transformation when using ξ .

3.3.2 Equilibrium on the nonlinear manifold

Learning the decoder specifies where the reduced solution may lie, but does not determine its location for a new input. For an iterative PROM-ANN, that location is selected by projected equilibrium. At fixed $\mathbf{e}$, the virtual displacement is $\delta \tilde{\mathbf{u}} = \mathbf{B}(\mathbf{q}) \delta \mathbf{q}$, giving

$$\mathbf{r}(\mathbf{q}; \mathbf{e}) = \mathbf{B}(\mathbf{q})^T \mathbf{R}(\tilde{\mathbf{u}}(\mathbf{q}); \mathbf{e}) = \mathbf{0}. \quad (28)$$

Its exact derivative is

$$\frac{\partial \mathbf{r}}{\partial \mathbf{q}} = \mathbf{B}^T \mathbf{K} \mathbf{B} + \sum_{j=1}^N R_j \frac{\partial^2 \tilde{u}_j}{\partial \mathbf{q} \partial \mathbf{q}}. \quad (29)$$

The second term accounts for manifold curvature. It need not vanish at reduced equilibrium, because $\mathbf{B}^T \mathbf{R} = \mathbf{0}$ does not imply $\mathbf{R} = \mathbf{0}$. Dropping it defines an approximate iteration matrix, not the exact Jacobian.

The latent closure does not prescribe the projection method. Least-squares Petrov–Galerkin (LSPG) instead minimizes a discrete residual norm and uses a test space involving the residual Jacobian and decoder tangent [18, 20, 43]. We use Galerkin virtual work for the present periodic hyperelastic problem. This choice retains the primary/secondary construction without identifying Galerkin and LSPG as the same reduced equations.

3.4 Hyperreduction of the projected equations

Although the PROMs reduce the number of equilibrium unknowns, evaluating their projected equations still requires full-mesh contributions. ECM addresses this remaining cost by replacing the element sum with a sparse weighted sum [29]. This is a second approximation, applied after choosing the displacement representation.

3.4.1 Fixed elementwise cubature

Let $\mathcal{E}$ be the microscopic element set and $\mathbf{L}_e$ extract element nodal values from $\mathbf{d}$. At fixed macroscopic input, define the reconstructed element state and its reduced tangent by

$$\mathbf{d}_e = \mathbf{L}_e(\mathbf{T}\tilde{\mathbf{u}} + \mathbf{g}_{\text{per}}), \quad \mathbf{B}_e = \mathbf{L}_e \mathbf{T} \mathbf{B}. \quad (30)$$

The full reduced residual is $\sum_{e \in \mathcal{E}} \mathbf{h}_e$, where $\mathbf{h}_e = \mathbf{B}_e^T \mathbf{f}_{\text{int},e}$. Fixed ECM approximates it as

$$\mathbf{r}_{\text{ECM}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e \mathbf{h}_e(\mathbf{q}; \mathbf{e}) = \mathbf{0}, \quad \mathcal{Z}_r \subset \mathcal{E}, \quad w_e \geq 0. \quad (31)$$

For the conventional HPROM, $\mathbf{B}$ is replaced by $\mathbf{V}_{\text{tra}}$ and the unknown is $\mathbf{q}_{\text{tra}}$. There is no ANN in this tier. A selected element retains its ordinary within-element quadrature. Consequently, selected elements, selected Gauss points, and independent coordinates are distinct counts.

To build the rule offline, we must retain the individual element contributions before summing them. Let $N_{\text{el}} = |\mathcal{E}|$, and let $\mathbf{c}_e^s \in \mathbb{R}^d$ collect the d integrand components of element e at training state s , for $s = 1, \dots, N_c$. For residual cubature, $\mathbf{c}_e^s = \mathbf{h}_e^s$, with $d = n_{\text{tra}}$ for the affine model and $d = n$ for the nonlinear manifold. Output cubature instead uses the four components of the element contribution to the average first Piola tensor:

$$\mathbf{p}_e^s = \frac{1}{|\mathbf{Y}|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu,g}^s, \quad \mathbf{c}_e^s = \text{vec } \mathbf{p}_e^s = [p_{e,11}^s \quad p_{e,12}^s \quad p_{e,21}^s \quad p_{e,22}^s]^T, \quad d = 4, \quad (32)$$

where ω_g is the reference quadrature measure, $\mathbf{P}_{\mu,g}^s$ is the microscopic first Piola stress, and vec uses the order (11, 12, 21, 22). Residual and output rules are built separately. Physical integration measures are already included in $\mathbf{c}_e^s$; the unknown cubature weights are element multipliers.

The raw integrand matrix arranges one element per row and one component of one training state per column:

$$\mathbf{M}_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \dots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \dots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} \in \mathbb{R}^{N_{\text{el}} \times (dN_c)}, \quad (\mathbf{M}_{\text{int}})_{e,(s-1)d+\alpha} = c_{e,\alpha}^s. \quad (33)$$

Thus the contribution vectors are concatenated across states, not assembled across elements. Compressing this matrix with a truncated SVD gives $\mathbf{M}_{\text{int}} \simeq \mathbf{U}_{\text{cub}} \mathbf{\Sigma}_{\text{cub}} \mathbf{V}_{\text{cub}}^T$, where $\mathbf{U}_{\text{cub}} \in \mathbb{R}^{N_{\text{el}} \times r_c}$ contains the r_c retained integration modes and $\mathbf{U}_{\text{cub}}^T \mathbf{U}_{\text{cub}} = \mathbf{I}$. These are modes of the *integrands*, not displacement POD modes.

Transposing $\mathbf{U}_{\text{cub}}$ makes each candidate element a column of the fitting system. The deployed ECM implementation adds the normalized part of the constant vector not already represented by that basis:

$$\mathbf{p}_{\perp} = (\mathbf{I} - \mathbf{U}_{\text{cub}} \mathbf{U}_{\text{cub}}^T) \mathbf{1}, \quad \mathbf{c}_{\perp} = \frac{\mathbf{p}_{\perp}}{\|\mathbf{p}_{\perp}\|_2}, \quad \mathbf{G}_{\text{cub}} = \begin{bmatrix} \mathbf{U}_{\text{cub}}^T \\ \mathbf{c}_{\perp} \end{bmatrix} \in \mathbb{R}^{(r_c+1) \times N_{\text{el}}}. \quad (34)$$

This expression assumes $\|\mathbf{p}_{\perp}\|_2 > 0$, as in the deployed rules; if the constant is already in the retained span, the additional constraint is redundant. Exact reproduction of these rows implies $\mathbf{1}^T \mathbf{w} = N_{\text{el}}$. Orthogonalizing the constant row avoids duplicating its component in the retained integration space. Figure 2(a) shows the passage from sampled contributions to this compressed system. The fitting problem on a selected support has the form

$$\min_{\mathbf{w} \geq 0} \|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2, \quad \mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}, \quad (35)$$

Here unselected weights are zero. Nonnegative least squares alone does not select a sparse rule: $\mathbf{w} = \mathbf{1}$ is the full-mesh solution. ECM combines fitting with greedy support selection until the relative integration residual satisfies

$$\frac{\|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2}{\|\mathbf{b}_{\text{cub}}\|_2} \leq \varepsilon_{\text{ECM}}. \quad (36)$$

The tolerance ε_{ECM} applies to the compressed integration constraints. It is distinct from the integrand-SVD truncation tolerance ε_{SVD} , which determines how many constraints are retained, and from the subsequent error

in an effective output. Training retains the individual contributions even when their assembled equilibrium sum vanishes. Further details of integration-basis compression and greedy selection are given in Ref. [29].

The deployed element-multiplier rules also impose $\sum_e w_e = |\mathcal{E}|$, with physical quadrature measures already contained in the element integrands. This is an element-count normalization, not by itself exact physical-volume integration on an unequal-volume mesh. Once constructed, both support and weights remain fixed online.

3.4.2 Manifold-adaptive weights

A single weight vector must satisfy integration constraints collected from every training state. Allowing the weights to depend on the reduced state relaxes that shared-weight requirement while retaining a common sparse support. This is the principle of MAW-ECM [32]. We use it with elementwise sampling and neural weight fields; the state representation remains the PROM-ANN decoder. The adaptive residual is

$$\mathbf{r}_{\text{MAW}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e(\mathbf{q}) \mathbf{B}_e(\mathbf{q})^T \mathbf{f}_{\text{int},e}(\mathbf{d}_e) = \mathbf{0}. \quad (37)$$

Fixed ECM first supplies a candidate support. Adaptive-weight pruning then removes elements while readjusting state-specific weights subject to the integration constraints and nonnegativity, until a prescribed target support size is reached. The neural fields are fitted on this pruned support, not on a fixed-ECM support selected directly at the final size. The pruning formulation, including its local and graph-coupled regularization mechanisms, is detailed in Ref. [32]; the available support-generation script activates the local positivity-enforcing phase from the first elimination. Its settings and the distinction between support selection and field fitting are recorded with the numerical example.

Pruning determines *which elements remain*; it does not yet provide a neural map for evaluating their weights at a new state. After pruning, we therefore train an additional ANN whose outputs determine the cubature weights. This is a separate network from the displacement closure $\mathcal{N}$, which has already been trained and remains fixed during weight fitting. Because equilibrium and stress use separate integration rules, the deployed model actually contains two such weight ANNs, denoted $\mathcal{N}_r$ and $\mathcal{N}_s$. Their roles are

$$\begin{aligned} \bar{\mathbf{q}}(\mathbf{q}) &= \mathcal{N}(\mathbf{q}; \boldsymbol{\eta}) \in \mathbb{R}^{\bar{n}}, & \text{secondary-coordinate ANN,} \\ \boldsymbol{\ell}_r(\mathbf{q}) &= \mathcal{N}_r(\mathbf{q}; \boldsymbol{\theta}_r) \in \mathbb{R}^{m_r}, & \text{residual-weight ANN,} \\ \boldsymbol{\ell}_s(\mathbf{q}) &= \mathcal{N}_s(\mathbf{q}; \boldsymbol{\theta}_s) \in \mathbb{R}^{m_s}, & \text{stress-weight ANN,} \end{aligned} \quad (38)$$

where $m_r = |\mathcal{Z}_r|$ and $m_s = |\mathcal{Z}_s|$. Thus the first ANN predicts secondary coordinates, not integration weights. Each additional ANN produces one scalar logit per element of its own fixed support. A prescribed output transformation converts these logits into positive element multipliers:

$$\begin{aligned} \mathbf{w}(\mathbf{q}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_r(\mathbf{q}; \boldsymbol{\theta}_r)), & \sum_{e \in \mathcal{Z}_r} w_e = N_{\text{el}}, \\ \mathbf{v}(\mathbf{q}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_s(\mathbf{q}; \boldsymbol{\theta}_s)), & \sum_{e \in \mathcal{Z}_s} v_e = N_{\text{el}}. \end{aligned} \quad (39)$$

Here $\text{softmax}(\boldsymbol{\ell})_j = \exp(\ell_j) / \sum_k \exp(\ell_k)$ acts over the selected elements, not over training states. Input standardization is absorbed into each network map. As for the closure, the implementation evaluates these maps in the strain-informed coordinates $\boldsymbol{\xi} = \mathbf{A}_m^{-1} \mathbf{q}$. The softmax enforces positivity and the prescribed sum; integration accuracy must still be learned. The offline sequence is therefore closure training, support selection and pruning, then training of the two weight ANNs. Details of the pruning algorithm are given by Hernández et al. [32]; the following equations specify our neural weight-field fit.

Once a common support $\mathcal{Z} = \{e_1, \dots, e_m\}$ has been selected, weight-field training returns to the uncompressed contributions at each decoded training state. With the integrands defined above, the statewise system is

$$\mathbf{A}^s = \begin{bmatrix} \mathbf{c}_{e_1}^s & \cdots & \mathbf{c}_{e_m}^s \\ 1 & \cdots & 1 \end{bmatrix} \in \mathbb{R}^{(d+1) \times m}, \quad \mathbf{b}^s = \begin{bmatrix} \sum_{e \in \mathcal{E}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix} \in \mathbb{R}^{d+1}. \quad (40)$$

Each vector $\mathbf{c}_{e_j}^s$ occupies d rows of one column. The first d rows reproduce the full-mesh integral; the last imposes the element-multiplier sum. In particular, $\mathbf{b}^s$ is computed from *all* elements, not by summing the selected columns of $\mathbf{A}^s$. Residual training uses $d = n$ and $\mathcal{Z} = \mathcal{Z}_r$; stress training uses $d = 4$ and its separate support $\mathcal{Z}_s$, replacing $\mathbf{w}$ by the stress multipliers $\mathbf{v}$.

For example, on a stress support of $m_s = |\mathcal{Z}_s|$ elements, the first four rows contain $p_{e,11}^s, p_{e,12}^s, p_{e,21}^s$, and $p_{e,22}^s$, respectively. Together with the row of ones, these form a $5 \times m_s$ matrix, denoted $\mathbf{A}_\sigma^s$ in Figure 2(c).

Its target contains the four components of the full-mesh average $\bar{\mathbf{P}}^s = \sum_{e \in \mathcal{E}} \mathbf{p}_e^s$, followed by N_{el} . There are four stress rows rather than three because first Piola stress is not generally symmetric; conversion to the three-component second Piola output is performed afterwards, as described in Section 3.5. For fixed stress cubature, the same four-component contributions instead populate $\mathbf{M}_{\text{int}} \in \mathbb{R}^{N_{\text{el}} \times 4N_c}$ and follow the compression and selection in panel (a).

Unlike $\mathbf{G}_{\text{cub}}$, $\mathbf{A}^s$ contains physical component rows for a single state rather than compressed integration modes over the training set. Its column identities remain fixed as s changes, while both their contributions and the predicted weights vary; see Figure 2(b). For the residual-weight ANN, only $\boldsymbol{\theta}_r$ is optimized in the integration loss

$$\mathcal{L}_{\text{cub}}(\boldsymbol{\theta}_r) = \frac{1}{|\mathcal{I}_{\text{fit}}|} \sum_{s \in \mathcal{I}_{\text{fit}}} \frac{\|\mathbf{A}^s \mathbf{w}(\mathbf{q}^s) - \mathbf{b}^s\|_2^2}{\max(\|\mathbf{b}^s\|_2^2, \delta_{\text{floor}})}. \quad (41)$$

The dependence on $\boldsymbol{\theta}_r$ enters through $\mathbf{w}$ in Eq. (39); the sampled matrices and targets remain fixed. Stress-weight training optimizes $\boldsymbol{\theta}_s$ in the analogous loss using $\mathbf{A}_\sigma^s$, $\mathbf{b}_\sigma^s$, and $\mathbf{v}$. The small positive floor avoids division by zero. A preliminary fit to per-state constrained weights initializes each network; the final objective concerns integration error, not agreement with a unique target weight vector. Different weight vectors can reproduce the same integral. Their mean-square training loss is distinct from the median relative integration error used for checkpoint selection or reporting.

Online, a reduced equilibrium iterate supplies the same current state to the closure ANN and the residual-weight ANN: one reconstructs displacement, while the other supplies the multipliers used to assemble its projected residual. The stress-weight ANN supplies the separate output rule. Network parameters and support identities remain fixed; only their evaluated outputs change with the state. No pruning or network training is performed inside the equilibrium iterations. In particular, the iterative model evaluates weights at the current reduced state, not by imposing $\boldsymbol{\xi} = \mathbf{e}$; that prescription belongs to the direct mode introduced below.

Differentiation gives

$$\mathbf{r}_{\text{MAW}, \mathbf{q}} = \sum_{e \in \mathcal{Z}_r} \left[w_e \mathbf{B}_e^T \mathbf{K}_e \mathbf{B}_e + w_e \sum_j (f_{\text{int}, e})_j \partial_{\mathbf{q}\mathbf{q}}^2 (d_e)_j + \mathbf{h}_e (\nabla_{\mathbf{q}} w_e)^T \right]. \quad (42)$$

Here $\mathbf{K}_e = \partial_{\mathbf{d}_e} \mathbf{f}_{\text{int}, e}$. The three terms arise from element stiffness, decoder curvature, and weight variation, respectively. The microscopic iteration uses only the projected-stiffness term as a modified-Newton matrix, while refreshing decoder values, tangents, and weights at each iterate. This approximation concerns the iteration matrix, not the definition of equilibrium. The constitutive tangent must instead differentiate the complete residual and output, as described next.

For the affine HPROM with fixed weights, both additional terms vanish: the decoder has zero curvature and the weights have zero state derivative. Its projected stiffness is therefore the exact residual Jacobian. The iterative comparison consequently uses exact reduced Newton for the affine model and modified Newton for HPROM-ANN. Differences in iteration count and derivative cost are part of the measured implementation-level speedup, not an isolated measure of the advantage of nonlinear coordinates.

3.5 Effective stress and consistent constitutive differentiation

Accurate integration of the projected residual does not automatically give accurate integration of the effective stress: these are different integrands. We therefore fit a separate output support $\mathcal{Z}_s$ and multipliers v_e , using fixed ECM for HPROM or the adaptive procedure of Section 3.4.2 for HPROM-ANN. In the following common expression, v_e is constant in the former case and depends on the reduced state in the latter. With microscopic first Piola stress $\mathbf{P}_{\mu, g}$ and reference quadrature measure ω_g , the stress reconstruction is

$$\bar{\mathbf{P}}_h(\mathbf{q}, \mathbf{e}) = \frac{1}{|Y|} \sum_{e \in \mathcal{Z}_s} v_e(\mathbf{q}) \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}(\mathbf{d}_e), \quad \bar{\mathbf{S}}_h = \bar{\mathbf{F}}^{-1} \bar{\mathbf{P}}_h. \quad (43)$$

Equation (43) provides the output counterpart of the residual rules in Eqs. (31) and (37). The residual and output supports may overlap. Their union specifies the element data required by an iterative constitutive evaluation, but only the residual support is used in equilibrium iterations.

The implemented three-component output takes the symmetric part of the converted tensor:

$$\mathbf{s}_h(\mathbf{q}, \mathbf{e}) = ((\bar{\mathbf{S}}_h)_{11}, (\bar{\mathbf{S}}_h)_{22}, \tfrac{1}{2}[(\bar{\mathbf{S}}_h)_{12} + (\bar{\mathbf{S}}_h)_{21}])^T. \quad (44)$$

(a) Fixed ECM: all training states, one weight vector

$$\begin{array}{ccc}
 \begin{array}{c} \text{Truncated SVD} \\ \text{transpose} \\ \text{add constraint} \end{array} & & \begin{array}{c} \text{Greedy} \\ \text{selection} \end{array} \\
 \begin{array}{c} M_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \cdots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \cdots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} \end{array} & \longrightarrow & \begin{array}{c} G_{\text{cub}} = \begin{bmatrix} U_{\text{cub}}^T \\ \mathbf{c}_{\perp}^T \end{bmatrix} \end{array} \\
 \begin{array}{c} \text{Rows: elements} \\ \text{Columns: state-component pairs} \\ N_{\text{el}} \times (dN_c) \end{array} & & \begin{array}{c} \text{Columns: candidate elements} \\ \text{Rows: modes} \\ \text{+ orthogonalized constant} \\ (r_c + 1) \times N_{\text{el}} \end{array} \\
 & & \begin{array}{c} G_{\text{cub}} \mathbf{w} \simeq \mathbf{b}_{\text{cub}}, \quad \mathbf{w} \geq \mathbf{0} \\ \mathbf{b}_{\text{cub}} = G_{\text{cub}} \mathbf{1} \end{array} \\
 & & \begin{array}{c} \text{One } \mathbf{w} \text{ for all retained modes} \\ w_e = 0 \text{ outside the support} \end{array}
 \end{array}$$

(b) MAW-ECM weight-field fit: one system per state, common support

$$\begin{array}{ccc}
 \begin{array}{c} \text{Columns: selected elements } e_1, \dots, e_m \\ \begin{bmatrix} \mathbf{c}_{e_1}^s & \cdots & \mathbf{c}_{e_m}^s \\ 1 & \cdots & 1 \end{bmatrix} \end{array} & \begin{array}{c} \begin{bmatrix} w_{e_1}(\mathbf{q}^s) \\ \vdots \\ w_{e_m}(\mathbf{q}^s) \end{bmatrix} \\ \underbrace{\hspace{1cm}}_{\mathbf{w}(\mathbf{q}^s)} \\ \text{State-dependent weights} \end{array} & \simeq & \begin{array}{c} \begin{bmatrix} \sum_{e \in \mathcal{E}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix} \\ \mathbf{b}^s: \text{full-mesh sum} \\ (d+1) \times 1 \end{array} \\
 \begin{array}{c} \mathbf{A}^s: d \text{ component rows + normalization} \\ (d+1) \times m \end{array} & & & &
 \end{array}$$

$$A^1 \mathbf{w}(\mathbf{q}^1) \simeq \mathbf{b}^1$$

...

$$A^{N_c} \mathbf{w}(\mathbf{q}^{N_c}) \simeq \mathbf{b}^{N_c}$$

After pruning: one weight ANN shared across all training states

$$\mathbf{q} \xrightarrow{\mathcal{N}_r} \ell_r \xrightarrow{N_{\text{el}} \text{ softmax}} \mathbf{w}(\mathbf{q})$$

(c) Adaptive stress example of (b): four first Piola components + normalization

$$\begin{array}{ccc}
 \mathbf{p}_e^s = \frac{1}{|Y|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}^s, & \bar{\mathbf{P}}^s = \sum_{e \in \mathcal{E}} \mathbf{p}_e^s, & \mathcal{Z}_s = \{e_1, \dots, e_{m_s}\} \\
 \begin{bmatrix} p_{e_1, 11}^s & \cdots & p_{e_{m_s}, 11}^s \\ p_{e_1, 12}^s & \cdots & p_{e_{m_s}, 12}^s \\ p_{e_1, 21}^s & \cdots & p_{e_{m_s}, 21}^s \\ p_{e_1, 22}^s & \cdots & p_{e_{m_s}, 22}^s \\ 1 & \cdots & 1 \end{bmatrix} & \begin{array}{c} \mathbf{v}(\mathbf{q}^s) \\ N_{\text{el}} \text{ softmax}(\mathcal{N}_s(\mathbf{q}^s)) \end{array} & \simeq & \begin{bmatrix} \bar{P}_{11}^s \\ \bar{P}_{12}^s \\ \bar{P}_{21}^s \\ \bar{P}_{22}^s \\ N_{\text{el}} \end{bmatrix} \\
 \mathbf{A}_{\sigma}^s \in \mathbb{R}^{5 \times m_s}: \text{selected contributions} & & & \mathbf{b}_{\sigma}^s \in \mathbb{R}^5: \text{full-mesh target}
 \end{array}$$

Fixed stress ECM: stack these contributions in $M_{\text{int}} \in \mathbb{R}^{N_{\text{el}} \times 4N_c}$, then follow panel (a).

Figure 2: Construction of the cubature fitting systems. (a) Fixed ECM compresses the sampled integrands and fits one nonnegative sparse weight vector; its last row is an orthogonalized constant. (b) Adaptive fitting uses uncompressed components, with one weight ANN and one support shared across states for each integration rule. (c) Stress instance of (b): four first Piola components and one normalization row, fitted with the stress multipliers $\mathbf{v}$. In (b)–(c), the scaled softmax enforces the last row by construction; the physical rows are fitted approximately. The prescribed sum N_{el} normalizes element multipliers, not physical volume. All targets use full-mesh sums. Residual and stress rules use separate supports and weight ANNs, distinct from the displacement-closure ANN.

This states the actual output convention without assuming that an approximate displacement and independently fitted stress cubature give an exactly symmetric raw tensor. The constitutive derivative refers to this output, including the symmetrization.

Once reduced equilibrium determines $\mathbf{q}^*(\mathbf{e})$, the effective map is $\mathbf{s}_h(\mathbf{q}^*(\mathbf{e}), \mathbf{e})$. If its residual Jacobian is nonsingular, the same implicit differentiation as in Section 2 gives

$$\mathbf{D}_h = \mathbf{s}_{h,e} - \mathbf{s}_{h,q} \mathbf{r}_{h,q}^{-1} \mathbf{r}_{h,e}. \quad (45)$$

Partial derivatives hold the other argument fixed and are evaluated at the converged state. In particular, $\mathbf{s}_{h,e}$ includes the periodic lifting and the first-to-second-Piola conversion. The residual derivative must be the derivative of the full residual, not merely the modified-Newton matrix used to reach equilibrium.

The implementations use the same implicit-function formula but divide its evaluation differently between analytic and numerical derivatives. For the affine HPROM, the assembled reduced stiffness and analytic lifting derivative give $d\mathbf{q}_{\text{tra}}^*/d\mathbf{e}$ through three linear right-hand sides. Central differences are then applied only to the stress, along these linearized equilibrium directions. For HPROM-ANN, the implementation instead uses central differences at fixed $\mathbf{q}$ or fixed $\mathbf{e}$ for all four partial-derivative matrices in Eq. (45), including the effects of decoder curvature and weight variation; a small linear solve combines them explicitly. These derivatives could also be obtained analytically or by automatic differentiation: numerical differentiation is an implementation choice, not a limitation of the formulation. Neither procedure re-solves nonlinear microscopic equilibrium at every perturbed strain, and both remain subject to differentiation and equilibrium-solver tolerances.

3.6 Direct decoder evaluation: D-HPROM-ANN

When input-informed coordinates closely approximate the loading parameters, the decoder can also be evaluated without solving reduced equilibrium, as discussed in the hyperelastic example of Ref. [32]. Here this defines a separate operating mode, D-HPROM-ANN, obtained by prescribing $\boldsymbol{\xi} = \mathbf{e}$ in Eq. (27):

$$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathbf{N}}(\mathbf{e}). \quad (46)$$

The decoded state is inserted directly into the adaptive stress rule, using the same closure and stress-weight field as the iterative model. Only the stress support is needed; no residual cubature or reduced equilibrium Jacobian is evaluated. The tangent is obtained by central differences of the complete direct stress map, updating the decoder, weights, lifting, and stress conversion at each perturbed input.

This route still evaluates the microscopic material law on selected elements; it is not a neural stress regressor or a learned macroscopic potential. The prefix D is our label for the direct operating mode, not a new projection method. Its accuracy depends on both the closure and the approximation $\boldsymbol{\xi} \approx \mathbf{e}$. Conversely, solving projected equilibrium corrects the primary state only within the chosen manifold and cubature approximation, so it does not guarantee a smaller stress error than direct evaluation at every input.

3.7 Retained mechanical structure and its limits

All of these reduced models evaluate the original microscopic hyperelastic law, and the iterative variants additionally impose projected equilibrium. These retained ingredients distinguish them from direct constitutive regression, but do not by themselves certify an effective energy potential.

With fixed weights and consistent kinematics, Galerkin forces derived from element energies are the gradient of a weighted reduced energy. This is the variational basis for structure-preservation results for ECSW under their stated hypotheses [28]. Transferring such a conclusion to the deployed effective stress map requires two additional distinctions.

First, the independently fitted equilibrium and stress cubatures need not define derivatives of one energy. Second, for element energies Π_e , weighting their derivatives differs from differentiating an energy with state-dependent weights:

$$\nabla_{\mathbf{q}} \sum_e w_e(\mathbf{q}) \Pi_e(\mathbf{q}, \mathbf{e}) = \sum_e w_e(\mathbf{q}) \nabla_{\mathbf{q}} \Pi_e + \sum_e \Pi_e \nabla_{\mathbf{q}} w_e. \quad (47)$$

The last term is absent from Eq. (37), and positivity of the weights does not remove it. Thus the weighted sum of element energies is not, in general, the potential of that residual. These observations do not assert failure at every state; they delimit what the construction guarantees. We assign neither an exact effective potential nor a polyconvexity certificate to the deployed HPROM variants without further analysis. A consistent tangent differentiates the implemented stress map; it does not make that map conservative or guarantee tangent symmetry.

3.8 Offline–online separation and reduced meshes

The offline stage collects HDM snapshots, constructs the ROB and latent coordinates, fits the closure, selects cubature supports, trains adaptive weights where applicable, and builds reduced meshes. The online stage evaluates the prescribed lifting, reconstructs only the displacement entries required by the active supports, solves reduced equilibrium when needed, and returns effective stress and tangent.

An optimized reduced mesh is constructed from the selected elements and their required nodes, preserving the original geometry, material law, and quadrature. The basis operators and periodic lifting, including its derivatives, are restricted to this support before online evaluation; the optimized implementations do not first construct a full-dimensional lifting vector. Cubature multipliers account approximately for omitted contributions. Element forces and stiffnesses are projected locally, and derivative evaluations are batched where possible. The direct mode also omits decoder derivatives unnecessary for stress evaluation.

Consequently, online cost depends on support size, retained secondary dimension, equilibrium iterations, and derivative evaluation, not only on the number of primary coordinates. Reducing the equilibrium dimension alone therefore does not determine the attainable speedup. Table 1 summarizes the deployed reduced models and their HDM reference; the full-mesh PROM constructions serve as intermediate formulations.

Table 1: Deployed reduced evaluators and their HDM reference. The independent displacement representations below are lifted to physical nodal fields through Eq. (13). Stress is reconstructed from microscopic constitutive evaluations.

Model	State representation	Equilibrium unknowns	Integration	Neural prediction
HDM / FOM	$\mathbf{u} \in \mathbb{R}^N$	N	Full mesh	None
HPROM	$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}$	n_{tra}	Fixed ECM	None
HPROM-ANN	$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q})$	n	Adaptive ECM	Secondary coordinates and weights
D-HPROM-ANN	$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e})$	None	Adaptive stress rule	Same closure and stress weights

4 Learnable paired-feature polyconvex energies

The non-intrusive route developed here replaces the repeated online solution of the microscopic problem by a learned effective stored energy. Its central constitutive relation is

$$\mathbf{s}(\mathbf{e}) = \nabla_{\mathbf{e}} W(\mathbf{e}), \quad (48)$$

where $\mathbf{s}$ is the stress work-conjugate to the engineering-strain vector $\mathbf{e}$. Thus all stress components derive from a common potential, and the Hessian of W supplies a major-symmetric material tangent. Fitting stress components independently does not, in general, enforce the cross-derivative identities required by such a potential [7]. This observation motivates the energy-based formulation adopted throughout this section.

One could stop at this point and construct a flexible hyperelastic network without restricting the curvature of its learned potential. We use precisely this option as the *Unconstrained energy* reference. It takes objective strain measures as inputs and obtains stress through Eq. (48). Beyond potential structure and objectivity, the only mechanical conditions imposed by construction are the reference conditions $W(\mathbf{I}) = 0$ and $\partial_{\mathbf{e}} W(\mathbf{0}) = \mathbf{0}$. The first fixes the arbitrary additive energy constant; the second is enforced by an affine correction that cancels the stress at the undeformed state, following the consistency construction of As’ad et al. [7].²

The Unconstrained energy model shows what can be represented before stronger finite-strain restrictions are introduced. Potential structure and reference normalization alone do not control deformation-gradient curvature or prevent finite energy as $J \rightarrow 0^+$. We therefore pursue polyconvexity by constructing the energy as a convex function of independent deformation minors, together with analytic terms that control the reference state and growth. The learned contribution combines a common family of paired directional features with one of two monotone convex cores: an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). These cores provide alternative representations while preserving the same feature construction and mechanical guarantees.

The remainder of this section makes these distinctions precise. We first state the mechanical requirements and separate potential structure from finite-strain stability. We then define the reference model, construct

²No convexity or growth condition is imposed on this reference.

the paired features and convex cores, and establish the properties of the complete energy. Throughout, $\mathbf{F} \in \text{GL}^+(2)$, $J = \det \mathbf{F} > 0$, and derivatives with respect to deformation are taken with the learned parameters fixed.

4.1 Admissibility criteria and scope of the guarantees

A small prediction error is not sufficient to define an admissible hyperelastic surrogate. We therefore separate six requirements relevant to the present construction. The requirements themselves are standard in constitutive PANN formulations [8, 9]; their grouping and labels below are specific to this paper:

- (C1) *Potential structure*: stress is the derivative of a single-valued stored energy in work-conjugate variables.
- (C2) *Objectivity*: a superposed rigid rotation does not change that energy.
- (C3) *Stress symmetry and anisotropy*: the material stress is symmetric, while directional dependence is allowed without imposing an unverified material symmetry group.
- (C4) *Reference normalization*: the undeformed state has zero energy and zero stress.
- (C5) *Polyconvexity*: the energy admits a convex representation in deformation minors, providing a sufficient rank-one convexity condition.
- (C6) *Growth*: energy diverges under volume collapse, unbounded volume expansion, and unbounded distortion at bounded positive volume.

This list is an organizing device, not a universal checklist or a claim that polyconvexity is necessary for every physical material. The Unconstrained energy reference satisfies (C1)–(C4) but carries no construction-level claim of (C5) or (C6). The ICNN and ICKAN models proposed here satisfy all six criteria by construction for every finite parameter set that obeys the stated architectural constraints.³

These statements rest on different levels of evidence. Constitutive identities follow from the representation itself; sufficient bounds hold under their stated assumptions; and sampled numerical checks verify an implementation but do not replace a proof. The discussion below maintains this distinction while examining each criterion.

Potential structure (C1). To ensure mutually compatible stresses, we derive all stress components from a single scalar potential. Equation (48) gives the resulting work-conjugate stress. It also implies that the work around every closed strain path Γ vanishes:

$$\oint_{\Gamma} \mathbf{s} \cdot d\mathbf{e} = 0. \quad (49)$$

For the isothermal purely elastic setting considered here, (C1) likewise gives $\mathbf{s} \cdot \dot{\mathbf{e}} - \dot{W} = 0$: the mechanical dissipation vanishes. Moreover, the stress Jacobian is the Hessian of W and therefore obeys the cross-derivative identities $\partial s_i / \partial e_j = \partial s_j / \partial e_i$. These statements follow from the scalar-potential representation itself, not from a small error at sampled states.

Objectivity and the three meanings of symmetry (C2)–(C3). Objectivity must remove dependence on the spatial observer without inadvertently imposing material isotropy. For a superposed proper rotation $\mathbf{Q}$, it requires

$$W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F}), \quad \mathbf{P}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\mathbf{P}(\mathbf{F}). \quad (50)$$

Using $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ as an energy input enforces the first identity, because a superposed spatial rotation leaves $\mathbf{C}$ unchanged; differentiating the objective energy then gives the stress transformation. Objectivity does not itself imply either potential structure or material isotropy. For the energy models considered here, the symmetry $\mathbf{S} = \mathbf{S}^T$ follows from differentiation with respect to symmetric strain. It implies a symmetric Cauchy stress $\boldsymbol{\sigma} = J^{-1} \mathbf{F} \mathbf{S} \mathbf{F}^T$, as required by angular-momentum balance in the present continuum.

Objectivity still permits anisotropy. Material symmetry is a different invariance: a right action $\mathbf{F} \mapsto \mathbf{F} \mathbf{Q}_m$ leaves the energy unchanged only when $\mathbf{Q}_m$ belongs to the material symmetry group. Section 4.3 will implement this distinction using directions attached to the reference material; a superposed spatial rotation acts on $\mathbf{F}$ from the left and does not alter the directional stretch measured in that reference. Finally, the constitutive tangent $\mathbf{D} = \partial \mathbf{s} / \partial \mathbf{e} \in \mathbb{R}^{3 \times 3}$ measures how the three stress components change with engineering

³Global nonnegative energy is not part of (C1)–(C6); the saved checkpoints are audited separately in Section 5.5 using the sufficient bound in Appendix C.

strain, with entries $D_{ij} = \partial s_i / \partial e_j$. For an energy-based law it is the Hessian $\partial_{ee}^2 W$. Its *major symmetry*, $\mathbf{D} = \mathbf{D}^T$, is a third property: it follows from a twice differentiable scalar energy, not merely from a symmetric stress tensor.

Reference normalization (C4). To make the learned energy consistent with the undeformed reference state, we require both zero energy and zero stress:

$$W(\mathbf{I}) = 0, \quad \mathbf{s}(\mathbf{0}) = \mathbf{0}, \quad (51)$$

where $\mathbf{s}(\mathbf{0}) = \mathbf{0}$ is equivalent to $\mathbf{P}(\mathbf{I}) = \mathbf{0}$ in the undeformed state. The first condition fixes an arbitrary additive energy constant. The second is a substantive stress condition, and the admissible correction depends on the model class. Section 4.2 subtracts the raw reference value and a term linear in $\mathbf{E}$ for the Unconstrained energy reference. That correction preserves its scalar potential, but an arbitrary linear function of $\mathbf{E}$ need not preserve polyconvexity in $\mathbf{F}$. Section 4.5 therefore uses a different correction for the proposed models: it is affine in independent deformation minors and thus preserves convexity in those variables.

Polyconvexity and finite-strain stability (C5). Potential structure alone does not control the energy curvature under changes in deformation. Direct convexity in $\mathbf{F}$ is generally too restrictive for objective finite-strain energies with singular-volume growth. Polyconvexity instead imposes convexity in independent deformation minors [12, 13]. In two dimensions,⁴ this means that the energy admits the representation

$$W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F}), \quad \mathcal{P} \text{ convex in its independent arguments.} \quad (52)$$

Here $\mathcal{P}$ is defined on the convex set $\mathbb{R}^{2 \times 2} \times (0, \infty)$, treating its matrix and scalar arguments as independent; only its evaluation in W sets the second argument equal to $\det \mathbf{F}$. The cofactor is linear in the entries of a 2×2 matrix, so it need not be introduced as an additional independent argument. Convexity of $\mathcal{P}$ implies that W cannot curve downward along an admissible rank-one perturbation of $\mathbf{F}$. Appendix A derives this consequence and states its differential form.

To distinguish the present construction from alternatives that impose convexity in strain space [7, 10], we emphasize that convexity in $\mathbf{C}$ or $\mathbf{E}$ does not imply polyconvexity. The reason is that $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$ depends quadratically on $\mathbf{F}$; Appendix A provides the corresponding derivation and an analytical counterexample.

Growth and the limits of the guarantees (C6). Polyconvexity alone does not prevent volume collapse or unbounded deformation. We therefore require $W \rightarrow \infty$ as $J \rightarrow 0^+$, as $J \rightarrow \infty$, and as $\|\mathbf{F}\| \rightarrow \infty$ while J remains bounded away from zero and infinity. These growth conditions contribute to coercivity but do not alone prove existence of a minimizer: the standard existence argument also requires admissible boundary data and finite-energy competitors [13]. Nor does a constitutive barrier guarantee convergence of an unconstrained Newton iteration. Architectural guarantees and numerical solver behavior are therefore reported separately.

4.2 Unconstrained energy reference

The most flexible energy model used in the comparison is a smooth neural network $\widehat{W}_\theta$ acting on

$$\mathbf{x}(\mathbf{e}) = (C_{11} - 1, C_{22} - 1, C_{12}, J - 1)^T. \quad (53)$$

Here $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $J = \sqrt{\det \mathbf{C}}$, so the four objective inputs are functions of the same three engineering-strain components; they do not introduce an additional state variable. Denote the resulting raw energy by $\widetilde{W}(\mathbf{e}) = \widehat{W}_\theta(\mathbf{x}(\mathbf{e}))$. A generic network does not satisfy the reference conditions automatically, so we define the normalized energy as

$$W_{\text{unc}}(\mathbf{e}) = \widetilde{W}(\mathbf{e}) - \widetilde{W}(\mathbf{0}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0}) \cdot \mathbf{e}. \quad (54)$$

At $\mathbf{e} = \mathbf{0}$, the first two terms cancel and the linear term vanishes, giving $W_{\text{unc}}(\mathbf{0}) = 0$. Differentiation gives $\partial_{\mathbf{e}} W_{\text{unc}}(\mathbf{e}) = \partial_{\mathbf{e}} \widetilde{W}(\mathbf{e}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0})$, and therefore zero reference stress. Because the correction is affine in $\mathbf{e}$, it does not change the Hessian of the raw network. Equation (54) thus fixes the reference energy and stress while leaving the learned curvature unrestricted.

Using components of $\mathbf{C}$ in a material reference basis ensures objectivity under superposed spatial rotations but does not impose isotropy: the network may depend differently on C_{11} , C_{22} , and C_{12} . A change of material reference axes must transform these tensor components and the represented law consistently. This model is used only as the flexible energy-based reference for interpreting the constrained ICNN and ICKAN results.

⁴The standard three-dimensional definition uses $(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F})$. The two-dimensional guarantee established here does not imply an admissible extension to arbitrary out-of-plane deformation.

4.3 Feature design: expressiveness and admissibility

The first design question is how to describe deformation without erasing the effective anisotropy of the RVE. Classical invariants provide the natural starting point. For a three-dimensional right Cauchy–Green tensor $\mathbf{C}_3$, the familiar first, second, and third principal invariants are

$$I_1^{(3)} = \text{tr } \mathbf{C}_3, \quad I_2^{(3)} = \frac{1}{2}[(\text{tr } \mathbf{C}_3)^2 - \text{tr}(\mathbf{C}_3^2)], \quad I_3^{(3)} = \det \mathbf{C}_3. \quad (55)$$

They describe principal stretches without recording their orientation relative to the material. An energy depending only on these scalars is therefore isotropic. In the present two-dimensional setting, the corresponding two independent principal invariants are $I_1 = \text{tr } \mathbf{C}$ and $\det \mathbf{C} = J^2$; a third independent principal invariant is not available.

To retain directional information, one can supplement the isotropic measures with a structural direction $\mathbf{d}$. The familiar anisotropic stretch invariant is

$$I_4(\mathbf{d}) = \mathbf{d}^T \mathbf{C} \mathbf{d} = \mathbf{C} : (\mathbf{d} \otimes \mathbf{d}) = \|\mathbf{F} \mathbf{d}\|^2, \quad \|\mathbf{d}\| = 1. \quad (56)$$

It measures squared stretch along a material direction and remains unchanged by a superposed spatial rotation. Such directional invariants are standard ingredients of anisotropic hyperelastic energies [33, 34]. Here $\mathbf{d}$ need not represent a physical fiber: learned directions provide objective scalar descriptors of the RVE’s effective anisotropy. In two dimensions, $\mathbf{d}$ is naturally accompanied by its perpendicular $\mathbf{d}^\perp$. For this orthonormal pair,

$$I_4(\mathbf{d}) + I_4(\mathbf{d}^\perp) = I_1, \quad \|\text{cof } \mathbf{F} \mathbf{d}\|^2 = I_4(\mathbf{d}^\perp). \quad (57)$$

The first identity shows that the pair resolves how I_1 is distributed between two perpendicular directions: their sum recovers I_1 , while their individual values retain directional information. The second shows that a cofactor-based directional measure reads the stretch in the perpendicular direction rather than introducing an independent third in-plane descriptor. These observations motivate the paired directional building block used below.

Objective directional descriptors solve only the first part of the design problem. Because the downstream core is constrained to be nondecreasing in every input, the feature coordinates also determine which deformation states the core must order. The feature map must therefore be designed together with that monotonicity constraint. Consider a feature vector $\mathbf{z}(\mathbf{F})$ and a scalar core H nondecreasing in each input. For any two deformation states,

$$\mathbf{z}(\mathbf{F}_B) \geq \mathbf{z}(\mathbf{F}_A) \text{ componentwise} \implies H(\mathbf{z}(\mathbf{F}_B)) \geq H(\mathbf{z}(\mathbf{F}_A)). \quad (58)$$

This implication holds regardless of the network’s width. If the complete constitutive energy is $W = H(\mathbf{z}) + W_{\text{corr}}$, where W_{corr} collects the analytic normalization and growth terms, the ordering constrains the neural contribution $W - W_{\text{corr}}$ rather than the complete energy. The feature map and analytic terms must therefore be considered together.

This observation explains why the choice of an admissible feature construction can matter more than the capacity of its convex core. Changing the number of features does not change the number of independent strains: the features remain functions of the same three engineering-strain components. They embed that state into variables on which the network is constrained to be monotone and convex. Additional features can enlarge the admissible response family, but their number alone does not prove completeness.

We therefore seek features that retain directional stretch, reshape the componentwise ordering seen by the monotone core, and remain convex in the independent variables $(\mathbf{F}, J)$. The paired construction below achieves these three objectives by coupling directional stretch to volume through admissible determinant exponents. A feature can consequently decrease along a path when its denominator grows sufficiently, even if a numerator increases. Its reference derivative is also controlled analytically, which will allow the reference stress to be removed without sacrificing convexity.

4.3.1 Paired directional features

Let m denote the number of scalar features. For each $i = 1, \dots, m$, let $\mathbf{d}_i = (\cos \theta_i, \sin \theta_i)^T$ and $\mathbf{d}_i^\perp = (-\sin \theta_i, \cos \theta_i)^T$. Define

$$\begin{aligned} z_i(\mathbf{F}, J) &= \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} + \frac{I_4(\mathbf{d}_i^\perp)^{q_i}}{q_i J^{c_i}} \\ &= \frac{\|\mathbf{F} \mathbf{d}_i\|^{2p_i}}{p_i J^{b_i}} + \frac{\|\mathbf{F} \mathbf{d}_i^\perp\|^{2q_i}}{q_i J^{c_i}}, \end{aligned} \quad (59)$$

where θ_i sets the orientation of the reference directions, p_i, q_i are the powers of the squared-stretch invariants, and b_i, c_i control each contribution's coupling to the area ratio J . These roles provide expressiveness, but the exponents cannot be chosen arbitrarily if the feature is to remain convex. They satisfy

$$p_i, q_i \geq \frac{1}{2}, \quad 0 \leq b_i \leq 2p_i - 1, \quad 0 \leq c_i \leq 2q_i - 1. \quad (60)$$

The lower bounds on p_i and q_i make the corresponding powers of the directional norms convex, while the upper bounds on b_i and c_i prevent their coupling with J from introducing a negative-curvature direction. Appendix B derives these conditions from the Hessian of a single stretch-determinant contribution. In the learned-feature variants, θ_i and the four exponent parameters are optimized, with p_i, q_i, b_i, c_i subject to Eq. (60); the fixed-feature controls of Section 5.2 hold the same initialized feature parameters constant while training the core. We use $\text{softplus}(x) = \log(1 + e^x)$ and $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$, whose ranges for $x \in \mathbb{R}$ are $(0, \infty)$ and $(0, 1)$, respectively. With unconstrained parameters $\hat{p}_i, \hat{q}_i, \hat{b}_i, \hat{c}_i \in \mathbb{R}$, set

$$\begin{aligned} p_i &= \frac{1}{2} + \text{softplus}(\hat{p}_i), & b_i &= (2p_i - 1) \text{sigmoid}(\hat{b}_i), \\ q_i &= \frac{1}{2} + \text{softplus}(\hat{q}_i), & c_i &= (2q_i - 1) \text{sigmoid}(\hat{c}_i). \end{aligned} \quad (61)$$

The angles require no admissibility bound: for any fixed θ_i , the maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$ remain linear in $\mathbf{F}$. The transformations in Eq. (61) preserve the exponent inequalities during training, so a penalty loss is not needed to enforce them. For finite real values of the unconstrained parameters, the bounds in Eq. (60) hold strictly; boundary values are approached as limits. Each z_i is one scalar feature containing two perpendicular contributions. The two contributions need not have equal exponents. Consequently, pairing perpendicular directions does not impose invariance under their interchange.

The construction was also designed to simplify reference normalization. Its paired and scaled contributions make the reference strain gradient of every feature isotropic. Consequently, the neural contribution has an isotropic reference stress, which a single scalar determinant correction can cancel in Section 4.5. The following result gathers the objectivity, convexity, and reference-state properties needed there.

Proposition 1. *Under the constraints in Eq. (60), each feature in Eq. (59) is objective and convex in the independent variables $(\mathbf{F}, J)$ on $J > 0$. Its value and engineering-strain gradient at the identity, evaluated with $J = \det \mathbf{F}$, are*

$$z_i^0 = \frac{1}{p_i} + \frac{1}{q_i}, \quad \partial_{\mathbf{e}} z_i|_{\mathbf{0}} = \rho_i(1, 1, 0)^T, \quad \rho_i = 2 - \frac{b_i}{p_i} - \frac{c_i}{q_i}. \quad (62)$$

The proof is given in Appendix B. The mechanism behind the reference gradient is that $1/p_i$ and $1/q_i$ cancel the stretch-power multipliers upon differentiation. The two perpendicular directional contributions then sum to $2\mathbf{I}$, while the determinant terms contribute $-(b_i/p_i + c_i/q_i)\mathbf{I}$. Thus ρ_i is the scalar amplitude of an isotropic reference gradient: its two normal components are equal and its shear component is zero. This reference property does not imply isotropy of the response away from the reference state.

The raw features are now mechanically admissible, but their numerical magnitudes may differ. Before passing them to the core, we therefore use $y_i = (z_i - z_i^{\text{cen}})/s_i$, where z_i^{cen} is a centering value and $s_i > 0$ is a scaling factor. Both are independent of deformation for any fixed parameter set. This positive affine conditioning preserves both convexity in independent $(\mathbf{F}, J)$ and the componentwise ordering required by the monotone core. The choices used in the constitutive assessments are specified in Section 5.2.

This feature design has a precise but limited purpose: it supplies objective, expressive, and admissible inputs to the constrained core. Learning the directions does not amount to identifying or enforcing a complete material symmetry group, and no completeness or universal-approximation theorem is asserted for a finite collection of these features followed by a monotone convex core.

4.4 Monotone ICNN and integrated-hat ICKAN cores

The feature construction of Section 4.3 resolves the input side of the constitutive design: each z_i is objective and convex in the independent variables $(\mathbf{F}, J)$. The remaining question is how a learned core may combine these inputs without undoing that convexity. The core receives the conditioned vector $\mathbf{y} = (y_1, \dots, y_m)^T$, with $y_i = (z_i - z_i^{\text{cen}})/s_i$. Because $s_i > 0$, this affine conditioning preserves the convexity and componentwise ordering of the physical features.

A convex core alone is not sufficient. Composition with the convex feature map preserves convexity only when the core is also componentwise nondecreasing [8, 9]. We enforce these same two properties in two different ways. The ICNN learns nonnegative linear maps and applies fixed softplus activations at its hidden nodes. The ICKAN instead learns a convex, nondecreasing scalar function on every edge and sums the incoming contributions at each node. The following definitions make these two mechanisms precise; Figure 3 then compares their multilayer architectures.

4.4.1 Monotone ICNN

Let L denote the number of hidden layers and $\mathbf{a}^{(\ell)}$ the vector of activations in layer ℓ , for $\ell = 1, \dots, L$. We write $\hat{H}(\mathbf{y})$ for the scalar neural core acting on the conditioned features $\mathbf{y}$; the complete energy additionally includes the reference-normalization and growth terms of Section 4.5. The deployed ICNN has the form

$$\begin{aligned}\mathbf{a}^{(1)} &= \phi(\mathbf{U}_1 \mathbf{y} + \mathbf{b}_1), \\ \mathbf{a}^{(\ell)} &= \phi(\mathbf{W}_\ell \mathbf{a}^{(\ell-1)} + \mathbf{U}_\ell \mathbf{y} + \mathbf{b}_\ell), \quad \ell = 2, \dots, L, \\ \hat{H}(\mathbf{y}) &= \mathbf{v}^T \mathbf{a}^{(L)} + \mathbf{w}^T \mathbf{y} + b_{\text{out}},\end{aligned}\tag{63}$$

where $\phi = \text{softplus}$ is applied componentwise. Its derivatives satisfy $\phi'(x) = \text{sigmoid}(x) > 0$ and $\phi''(x) = \text{sigmoid}(x)[1 - \text{sigmoid}(x)] > 0$. Thus the activation is smooth, increasing, and convex; its smoothness also makes the core C^2 , as required for a continuous constitutive tangent. The matrices $\mathbf{U}_\ell$ and $\mathbf{W}_\ell$ contain the weights from $\mathbf{y}$ and the preceding hidden layer, respectively, and $\mathbf{b}_\ell$ is the layer bias vector. The output weight vectors $\mathbf{v}$ and $\mathbf{w}$ act on $\mathbf{a}^{(L)}$ and $\mathbf{y}$, respectively; b_{out} is an additive scalar output bias.

Every entry of $\mathbf{W}_\ell, \mathbf{U}_\ell, \mathbf{v}, \mathbf{w}$ is nonnegative, enforced by softplus transformations of unconstrained weight parameters. These signs play two related but distinct roles. The matrices $\mathbf{W}_\ell$ and the output vector $\mathbf{v}$ multiply convex hidden activations, so their nonnegativity preserves convexity as well as monotonicity. The maps $\mathbf{U}_\ell \mathbf{y}$ and $\mathbf{w}^T \mathbf{y}$ are affine and therefore do not affect convexity for either sign, but nonnegative $\mathbf{U}_\ell$ and $\mathbf{w}$ are required for componentwise monotonicity. Biases are unrestricted because adding constants to a preactivation preserves both properties. Hidden biases can change the core's derivatives with respect to $\mathbf{y}$, whereas b_{out} only shifts $\hat{H}$ and leaves its derivatives unchanged.

The preservation argument proceeds layer by layer. The first hidden layer is convex because softplus is convex and its argument is affine, and it is nondecreasing because $\mathbf{U}_1$ is nonnegative. Assuming that $\mathbf{a}^{(\ell-1)}$ is convex and nondecreasing, the nonnegative map $\mathbf{W}_\ell \mathbf{a}^{(\ell-1)}$, the nonnegative affine input map, and composition with ϕ preserve both properties in layer ℓ . The same argument applies to the output, proving by induction that $\hat{H}$ is convex and componentwise nondecreasing. This is the ICNN composition argument of Amos et al. [35], supplemented here by nonnegative direct-input maps to guarantee monotonicity.

4.4.2 Monotone integrated-hat ICKAN

Unlike the ICNN, which applies a fixed activation after an affine weighted sum at each hidden node, the ICKAN assigns a learned scalar function to each connection (edge) between layers. Each node sums the transformed contributions from its incoming connections. To make the relation between these edge functions and the scalar core explicit, set $\mathbf{h}^{(0)} = \mathbf{y}$ and let n_ℓ be the width of layer ℓ , with $n_0 = m$. The hidden layers and scalar output are

$$\begin{aligned} h_j^{(\ell)} &= \sum_{i=1}^{n_{\ell-1}} f_{ji}^{(\ell)}(h_i^{(\ell-1)}) + b_j^{(\ell)}, & j = 1, \dots, n_\ell, \quad \ell = 1, \dots, L, \\ \hat{H}(\mathbf{y}) &= \sum_{i=1}^{n_L} f_i^{\text{out}}(h_i^{(L)}) + \mathbf{w}^T \mathbf{y} + b_{\text{out}}, & \mathbf{w} \geq \mathbf{0}. \end{aligned} \quad (64)$$

Thus $f_{ji}^{(\ell)}$ transforms one scalar carried by an edge, $h_j^{(\ell)}$ is the sum formed at a node, and $\hat{H}(\mathbf{y})$ is the complete ICKAN core. The nonnegative linear skip $\mathbf{w}^T \mathbf{y}$ connects the conditioned input directly to the scalar output. It remains to specify the functions appearing on the edges. For any $f_{ji}^{(\ell)}$ or f_i^{out} in Eq. (64), let x denote its scalar input. Each edge uses the common form

$$f(x) = ax + \sum_k c_k B_k(x), \quad a, c_k \geq 0, \quad (65)$$

where a is the learned linear slope, c_k are learned weights of the spline bases B_k , and k indexes these bases. Softplus transformations of unconstrained parameters impose the signs of a and c_k . Additive constants are represented once, through the unrestricted node biases in Eq. (64), rather than redundantly on every incoming edge.

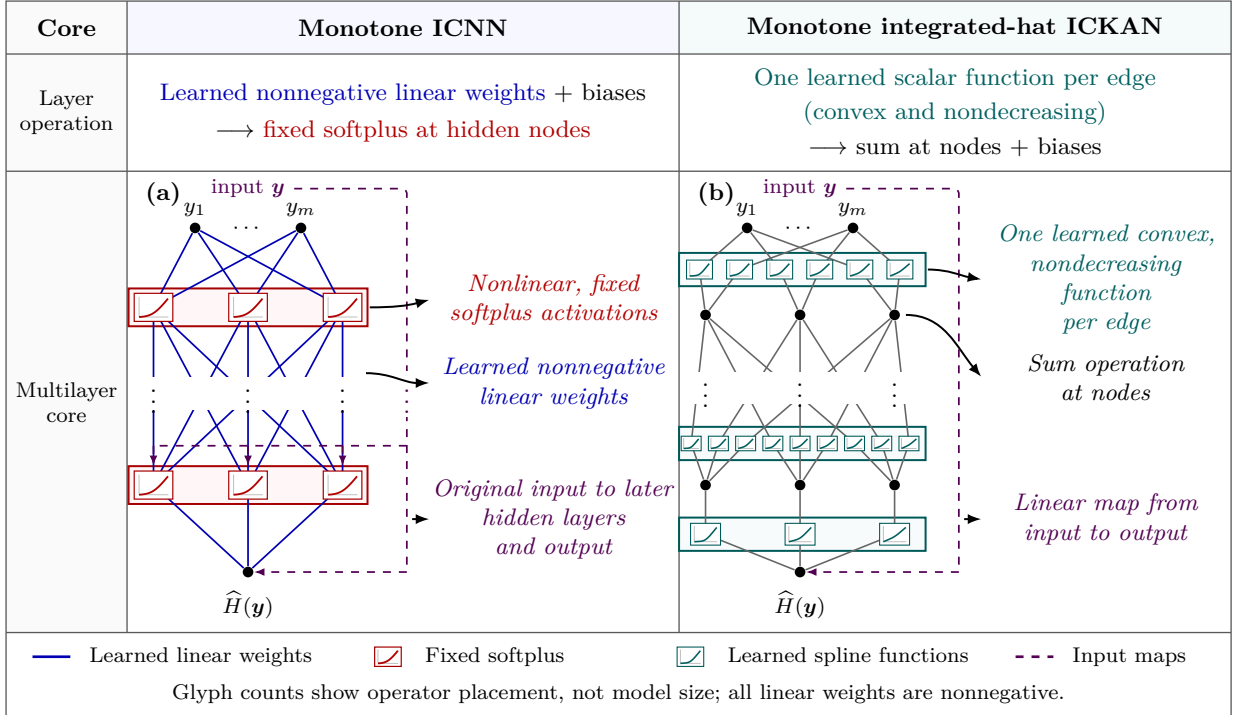


Figure 3: Multilayer monotone cores mapping the conditioned paired features $\mathbf{y}$ to the scalar contribution $\hat{H}(\mathbf{y})$. (a) The ICNN learns nonnegative linear maps and applies fixed softplus activations at hidden nodes. (b) The ICKAN learns one convex, nondecreasing scalar function per edge and sums them at nodes. Purple dashed paths denote nonnegative maps of the complete input; ellipses denote intervening layers. Glyph counts indicate where each operation acts, not deployed widths or parameter counts. Unrestricted biases are omitted, and the diagram represents the neural cores rather than the complete energy.

The remaining design task is to learn the shape of each edge without allowing negative slope or curvature. Rather than check these inequalities at sampled inputs after training, we parameterize the edge curvature directly. Each B_k'' is a nonnegative triangular hat centered at a knot κ_k . The knots form a fixed uniform

grid with spacing $h > 0$; neither their locations nor the spacing are optimized. Writing $s = (x - \kappa_k)/h$, one curvature basis is

$$B_k''(x) = \max(1 - |s|, 0). \quad (66)$$

Integrating twice from the left conditions $B_k(x) = B_k'(x) = 0$ for $x \leq \kappa_k - h$ gives

$$B_k(x) = h^2 \begin{cases} 0, & s \leq -1, \\ (s+1)^3/6, & -1 < s \leq 0, \\ (-s^3 + 3s^2 + 3s + 1)/6, & 0 < s < 1, \\ s, & s \geq 1. \end{cases} \quad (67)$$

Figure 4 shows both levels of the construction: integration of one curvature basis and assembly of a learned edge from several bases.

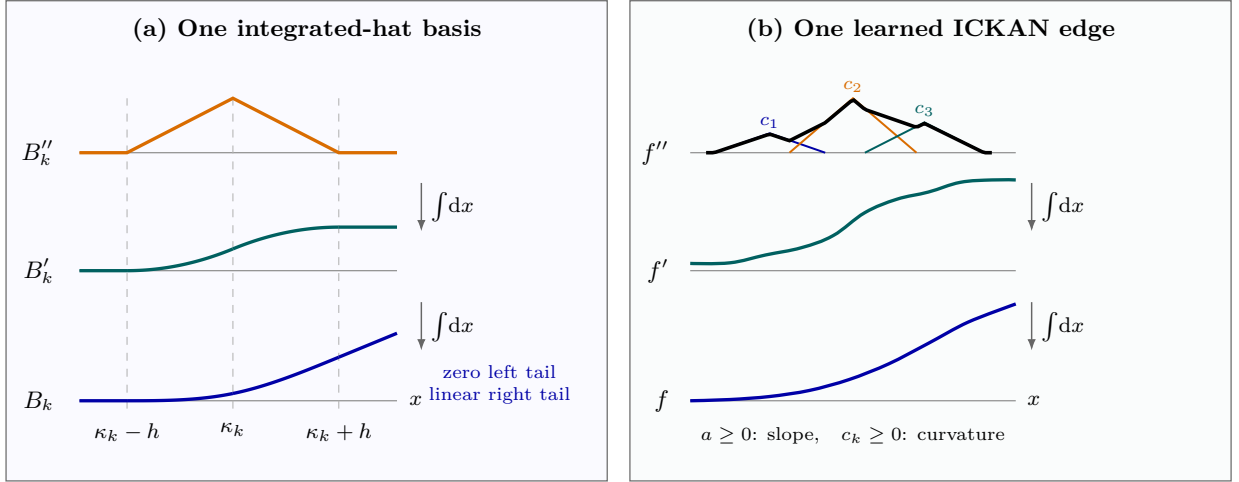


Figure 4: Integrated-hat parameterization of an ICKAN edge. (a) A localized nonnegative curvature hat integrates to a nonnegative slope and a convex C^2 basis with a zero left tail and a linear right tail. (b) Nonnegative learned coefficients combine several fixed bases into a nonnegative edge curvature; integration gives a nondecreasing slope and a convex, nondecreasing connection. The curves illustrate the analytic construction; they do not represent the deployed grid or parameter count.

The value and first two derivatives in Eq. (67) match at every join, and B_k'' vanishes at the outer joins. Thus B_k is globally C^2 . Its curvature is localized to $[\kappa_k - h, \kappa_k + h]$, but the integrated basis is not: it has a zero left tail and a linear right tail. Moreover, $B_k'(-\infty) = 0$ and $B_k'' \geq 0$ imply $B_k' \geq 0$. The complete edge therefore satisfies

$$f''(x) = \sum_k c_k B_k''(x) \geq 0, \quad f'(x) = a + \sum_k c_k B_k'(x) \geq 0, \quad (68)$$

for every learned parameter set with $a, c_k \geq 0$. Hence each edge is convex and nondecreasing by construction, rather than by a sampled numerical test.

Equation (64) sums these scalar edge functions at each node. Because every edge is convex and non-decreasing, induction across the layers shows that every $h_j^{(\ell)}$, and finally $\hat{H}$, is convex and componentwise nondecreasing in $\mathbf{y}$. The nonnegative input-to-output skip preserves both properties. The fixed knot grid and finite number of bases limit where an individual edge can learn curvature: outside their combined support, the edge has linear tails. These tails are linear in the scalar input x , not in $\mathbf{F}$; the complete energy still contains nonlinear features and analytic growth terms. The grid used in the constitutive assessment is reported in Section 5.2. This core is inspired by the monotonic input-convex KAN framework [41], with the integrated-hat bases and analytic tails specified here.

4.5 The complete energy and its admissibility guarantees

Sections 4.3 and 4.4 constructed the physical feature vector $\mathbf{z}$, its conditioned representation $\mathbf{y}(\mathbf{z})$, and the scalar neural core $\hat{H}(\mathbf{y})$. We write

$$H(\mathbf{z}) = \hat{H}(\mathbf{y}(\mathbf{z})) \quad (69)$$

for the resulting learned contribution. This term represents the anisotropic response, but it does not by itself set the energy and stress references or prescribe growth at the boundary of the admissible deformation domain.

To complete the construction, we first quantify the residual stress produced by H at the undeformed reference state. Let $\mathbf{z}^0 = \mathbf{z}(\mathbf{I}, 1)$. With the feature-gradient factors ρ_i from Eq. (62), define the core sensitivities and the scalar amplitude of this residual stress as

$$h_i = \partial_{z_i} H(\mathbf{z}^0) \geq 0, \quad r = \sum_i h_i \rho_i. \quad (70)$$

For a fixed parameter set, $H(\mathbf{z}^0)$ and r are constants with respect to deformation. The complete stored energy then combines the learned contribution with analytic reference-normalization and growth terms:

$$\begin{aligned} W(\mathbf{F}) = & \underbrace{H(\mathbf{z}) - H(\mathbf{z}^0)}_{\text{(i) } W_{\text{learn}}} + \underbrace{[-r(J-1)]}_{\text{(ii) } W_{\text{corr}}} \\ & + \underbrace{\alpha(J-1 - \log J) + \frac{\beta}{2}(J-1)^2}_{\text{(iii) } W_{\text{vol}}} \\ & + \underbrace{\varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right)}_{\text{(iv) } W_{\text{dist}}}, \quad \alpha, \beta, \varepsilon > 0. \end{aligned} \quad (71)$$

Here $J = \det \mathbf{F}$, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $\mathbf{z} = \mathbf{z}(\mathbf{F}, J)$. The positive energy-density coefficients α , β , and ε weight the analytic growth terms. For either core, the composition in Eq. (69) is twice continuously differentiable, convex, and componentwise nondecreasing in $\mathbf{z}$. The underbraces name the four contributions used below: W_{learn} carries the learned anisotropic response, while W_{corr} , W_{vol} , and W_{dist} impose the reference and growth behavior.

Each contribution solves a distinct constitutive problem:

- (i) **Learned response and energy reference (W_{learn}).** At $\mathbf{F} = \mathbf{I}$, the features equal $\mathbf{z}^0$, and hence

$$W(\mathbf{I}) = H(\mathbf{z}^0) - H(\mathbf{z}^0) = 0. \quad (72)$$

The remaining terms also vanish because $J = 1$ and $\text{tr } \mathbf{C} = 2$. Since $H(\mathbf{z}^0)$ is constant with respect to deformation, its subtraction fixes the energy reference without changing the stress or tangent.

- (ii) **Reference-stress correction (W_{corr}).** A constant subtraction cannot change stress. By the chain rule and Eq. (62), the learned contribution instead has the reference value

$$\nabla_{\mathbf{e}} H(\mathbf{z})|_0 = \sum_i h_i \nabla_{\mathbf{e}} z_i|_0 = \sum_i h_i \rho_i (1, 1, 0)^T = r(1, 1, 0)^T. \quad (73)$$

The identity $J = \sqrt{(1 + 2E_{11})(1 + 2E_{22}) - (2E_{12})^2}$ gives $\nabla_{\mathbf{e}} J|_0 = (1, 1, 0)^T$. Holding r fixed when differentiating therefore yields

$$\nabla_{\mathbf{e}} [H(\mathbf{z}) - r(J-1)]|_0 = r(1, 1, 0)^T - r(1, 1, 0)^T = \mathbf{0}. \quad (74)$$

A single scalar r suffices because the paired features make only the *reference* stress isotropic; the learned response away from $\mathbf{F} = \mathbf{I}$ remains anisotropic. The correction is zero in value at $J = 1$ and, being affine in the independent determinant variable, preserves convexity for either sign of r . This affine determinant normalization follows the strategy of Linden et al. [9]; Eq. (62) makes it applicable to the present directional feature family.

- (iii) **Volumetric growth (W_{vol}).** Its two parts control opposite limits: $\alpha(J-1 - \log J)$ diverges as $J \rightarrow 0^+$, whereas $\beta(J-1)^2/2$ grows quadratically as $J \rightarrow \infty$. At the reference,

$$W_{\text{vol}}(1) = 0, \quad W'_{\text{vol}}(J) = \alpha(1 - J^{-1}) + \beta(J-1), \quad W'_{\text{vol}}(1) = 0. \quad (75)$$

Thus this term changes neither the reference energy nor the reference stress. Both parts are needed: the quadratic term alone would remain finite at collapse, with limit $\beta/2$.

- (iv) **Distortional growth at bounded area** (W_{dist}). A function of J alone cannot detect a change of shape that preserves area. For example, with $\lambda > 0$,

$$\mathbf{F} = \text{diag}(\lambda, \lambda^{-1}), \quad J = 1, \quad \varepsilon \left(\frac{\text{tr} \mathbf{C}}{2} - 1 - \log J \right) = \frac{\varepsilon}{2} (\lambda^2 + \lambda^{-2} - 2). \quad (76)$$

This contribution diverges as $\lambda \rightarrow \infty$, although the area remains constant. More generally, $\text{tr} \mathbf{C} = \|\mathbf{F}\|_F^2$ supplies growth under unbounded deformation at bounded positive J . At the reference, the constant -1 sets its value to zero. Using $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$, its stress contribution is

$$\partial_{\mathbf{E}} \left[\varepsilon \left(\frac{\text{tr} \mathbf{C}}{2} - 1 - \log J \right) \right] \Big|_0 = \varepsilon [\mathbf{I} - J^{-1} \partial_{\mathbf{E}} J|_0] = \mathbf{0}. \quad (77)$$

Together, W_{learn} and W_{corr} set the learned response and its reference state, while W_{vol} and W_{dist} prescribe growth outside the sampled regime. Zero value and zero first derivative at the reference do not imply zero second derivative: the analytic terms still contribute to the reference stiffness because J is nonlinear in $\mathbf{e}$. The following proposition gathers the guarantees of their complete sum.

Proposition 2. *Let the features satisfy Eq. (60), let the core be either of the C^2 convex, componentwise nondecreasing constructions in Section 4.4, and let $\alpha, \beta, \varepsilon > 0$. Then Eq. (71) defines a C^2 , objective, two-dimensional polyconvex energy on $\text{GL}^+(2)$. Its work-conjugate stress $\mathbf{s} = \nabla_{\mathbf{e}} W$ is symmetric, $W(\mathbf{I}) = 0$, $\mathbf{s}(\mathbf{0}) = \mathbf{0}$, and W diverges under collapse, unbounded expansion, and unbounded distortion at bounded positive area. Thus the construction satisfies (C1)–(C6).*

The proof is given in Appendix B.3. In brief, the convex, componentwise nondecreasing core preserves feature convexity; the remaining terms are constant, affine, or convex in independent $(\mathbf{F}, J)$. The reference identities and growth limits then complete the argument. These guarantees hold for every admissible parameter set and therefore do not depend on the training samples or fitting error. Predictive accuracy and structural deployment are assessed separately in Sections 5.3 and 6. Global nonnegative energy is not an architectural guarantee; it is audited for the saved checkpoints in Section 5.5 using the sufficient bound derived in Appendix C.

The three constitutive constructions carried into the numerical assessment are summarized in Table 2. The ICNN and ICKAN columns include both fixed- and learned-feature variants; learning the feature parameters changes the representation, not the guarantees established above.

Table 2: Constitutive constructions carried into the numerical assessment. Criteria (C1)–(C6) are defined in Section 4.1; guarantees refer to the specific constructions in this section.

	Unconstrained energy	ICNN	ICKAN
Role	Flexible hyperelastic reference	Guaranteed paired-feature energy	Guaranteed paired-feature energy
Inputs	Objective measures from $(\mathbf{C}, J)$	Paired features (fixed or learned)	Paired features (fixed or learned)
Learned scalar map	Smooth energy network	Nonnegative weights; fixed activations	Convex, nondecreasing edge functions
(C1)–(C4)	By construction	By construction	By construction
(C5)–(C6)	Not imposed	By construction	By construction

Section 5 now evaluates two separate questions: whether the implementations reproduce these constructions and how accurately the resulting energies approximate the RVE response.

5 Constitutive assessment on periodic microstructures

In this section, we evaluate the proposed constitutive modeling approach on two periodic representative volume elements (RVEs) with different directional microstructures. Material A contains one elliptical cavity rotated relative to the cell axes; material B contains four elliptical cavities of different sizes and orientations. Both use the same isotropic matrix law and 20% porosity. Together they let us assess the learned energy in a cell with one prominent geometric direction and in a cell with several. We compare the surrogates with

independent full-order RVE responses and, in material B, assess the effect of learning the paired-feature parameters against fixed-feature counterparts. The material-A models are subsequently deployed in the structural FE² calculation of Section 6. We distinguish the guarantees established by construction from the numerical accuracy and consistency checks reported here.

5.1 Microstructures and reference-model verification

5.1.1 Material A: single-cavity reference

The square periodic cell contains a centered elliptical void with area fraction 0.20, aspect ratio 2, and major-axis angle 30° relative to the cell axes; see Fig. 5. This orientation avoids the mirror-aligned configurations of the square cell. Centrosymmetry remains, but does not mean that the strain energy is an even function of strain.

The solid is compressible Neo-Hookean in plane strain. With $\mathbf{C}_\mu = \mathbf{F}_\mu^T \mathbf{F}_\mu$ and $J_\mu = \det \mathbf{F}_\mu > 0$, its local energy density is

$$\psi_\mu = \frac{\mu}{2}(\text{tr } \mathbf{C}_\mu - 2) - \mu \log J_\mu + \frac{\lambda}{2}(\log J_\mu)^2, \quad (78)$$

with Young’s modulus 1.628 GPa and Poisson ratio 0.4. Here μ and λ are the usual Lamé constants. This is the specific matrix law used by the reference solver, not a generic assertion about every Neo-Hookean variant. The production mesh contains 1546 quadratic triangles, 3240 nodes, and 4638 quadrature points. Periodicity and null-space constraints leave 6320 independent microscopic displacement coordinates. Cell averages use the complete reference cell measure.

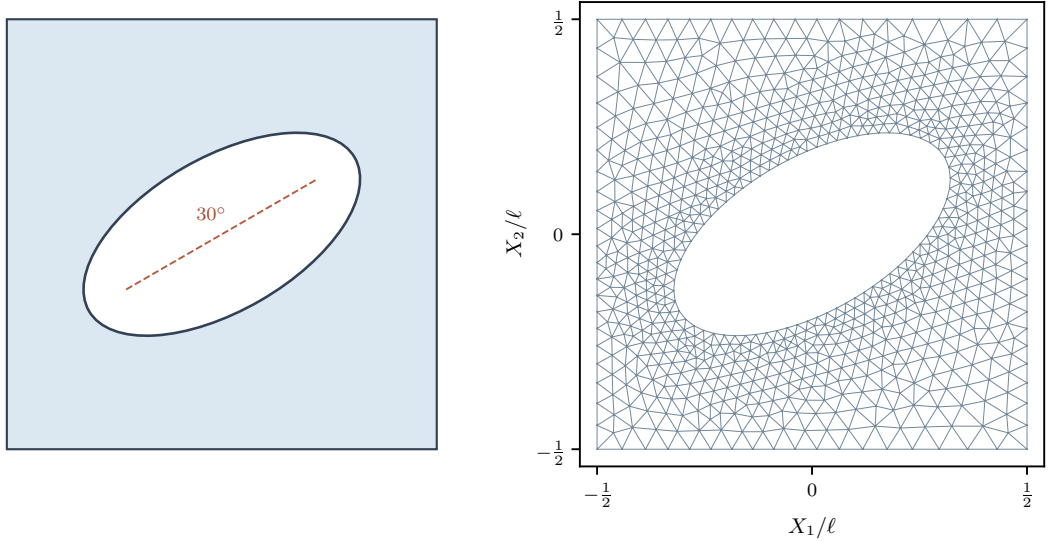


Figure 5: Material A: cavity geometry (left) and working quadratic mesh (right). Dashed lines mark cavity major axes; signed angles are measured counterclockwise from X_1 . Only edges joining corner nodes are drawn, and ℓ denotes the cell side.

The production mesh was selected using a local stress-field criterion along a uniaxial-stress path, rather than only a homogenized average. On this path, E_{22} and $2E_{12}$ are solved so that $\bar{S}_{22} = \bar{S}_{12} = 0$. At $E_{11} = 0.20$, its maximum microscopic first Piola stress norm differs by approximately 0.486% from a 3095-element reference, whereas the average axial stress differs by approximately 0.001%. This is a mesh check at a specified state, not a uniform error bound over all multiaxial data.

5.1.2 Material B: multicavity reference

Material B retains the square periodic cell, the plane-strain Neo-Hookean matrix law and moduli, and the 20% total void fraction of material A. The void space is instead distributed among four ellipses with unequal sizes, aspect ratios, orientations, and off-center positions; see Fig. 6. Holding the matrix and porosity fixed isolates the change in cavity arrangement. Unlike the single rotated void in A, this cell presents several geometric directions within the same microstructure. We do not assign a material symmetry group to the resulting effective law.

The working mesh contains 4621 quadratic triangles and 9582 nodes. Periodic boundary conditions and full-cell averaging follow Section 2.

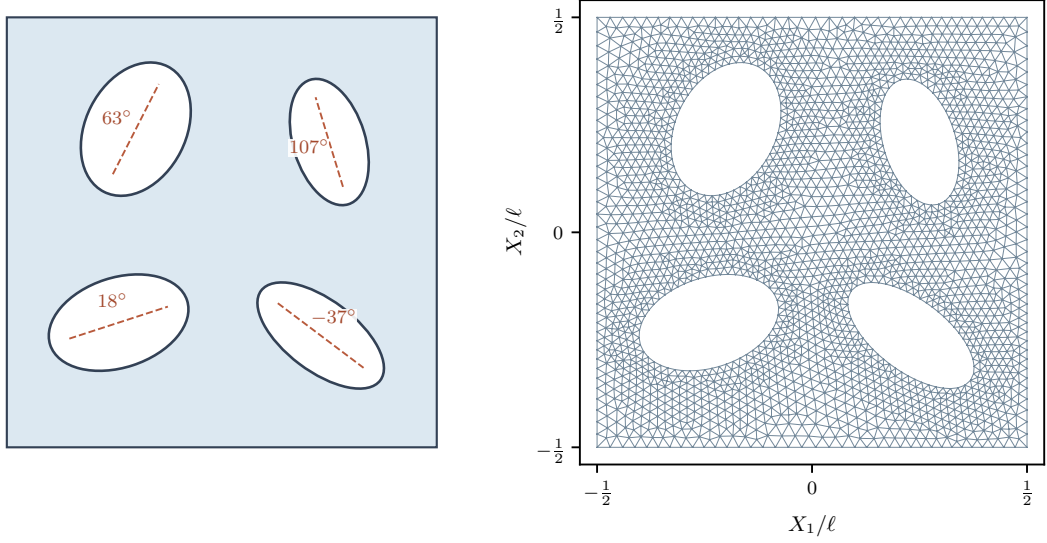


Figure 6: Material B: cavity geometry (left) and working quadratic mesh (right). Dashed lines mark cavity major axes; signed angles are measured counterclockwise from X_1 . Only edges joining corner nodes are drawn, and ℓ denotes the cell side.

At 64 prescribed multiaxial states, the working mesh was compared with an 8961-element mesh. Relative differences in homogenized energy, stress, and tangent remained below 0.1%; the sampled maximum microscopic first Piola stress norm differed by at most 4.06%. These finite comparisons support the working mesh, not a uniform convergence bound.

5.2 Data domains and learning protocol

Training domains and data roles. To define the training domain for material A, we first consider the problem it must solve. In Section 6, its learned law supplies the constitutive response throughout a coupon FE^2 calculation, where different macroscopic quadrature points visit different strains as loading proceeds. To locate those strains before training, we ran a displacement-controlled coupon prepass with a St Venant–Kirchhoff approximation based on the cell’s reference tangent. The prepass recorded the strain at every macroscopic quadrature point; it defined the sampling envelope, not the stresses used to train the models. Displacement control prevented errors in the approximate force response from determining that envelope. A second prepass with a softer response shifted the observed componentwise extrema by at most 5.1%; we expanded the spans by 40% to allow for such differences. Because the coupon is loaded in tension, we set the lower E_{11} bound to zero. In the engineering-strain coordinates $\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T$, the rounded bounds are

$$\mathbf{e}_{\min} = (0, -0.09865, -0.16807)^T, \quad \mathbf{e}_{\max} = (0.1955, 0.01635, 0.10793)^T. \quad (79)$$

We sample this box on a Cartesian grid, which also provides neighboring states for the reduced models of Section 3. Of its 4950 nonreference states, 4208 are used for fitting and 742 for validation (split seed 5). An independent set of 400 off-grid states tests interpolation.

Material B serves a different purpose: it tests the constitutive models on a cell with several geometric directions and permits a controlled comparison of fixed and learned features. It is not used in the coupon FE^2 calculation or a reduced model, so its sampling domain need not inherit A’s coupon-derived bounds or Cartesian grid. After full-order and mesh checks at multiaxial boundary states, we chose $E_{11}, E_{22} \in [-0.04, 0.20]$ and $2E_{12} \in [-0.08, 0.08]$. This box includes moderate normal compression as well as extension and shear within the checked range. Low-discrepancy samples in its interior and on its faces and corners supply 4200 fitting states; separate samplings supply 512 validation and 512 test states. Ten prescribed loading paths are reserved for evaluation. Figure 7 shows both sampling designs.

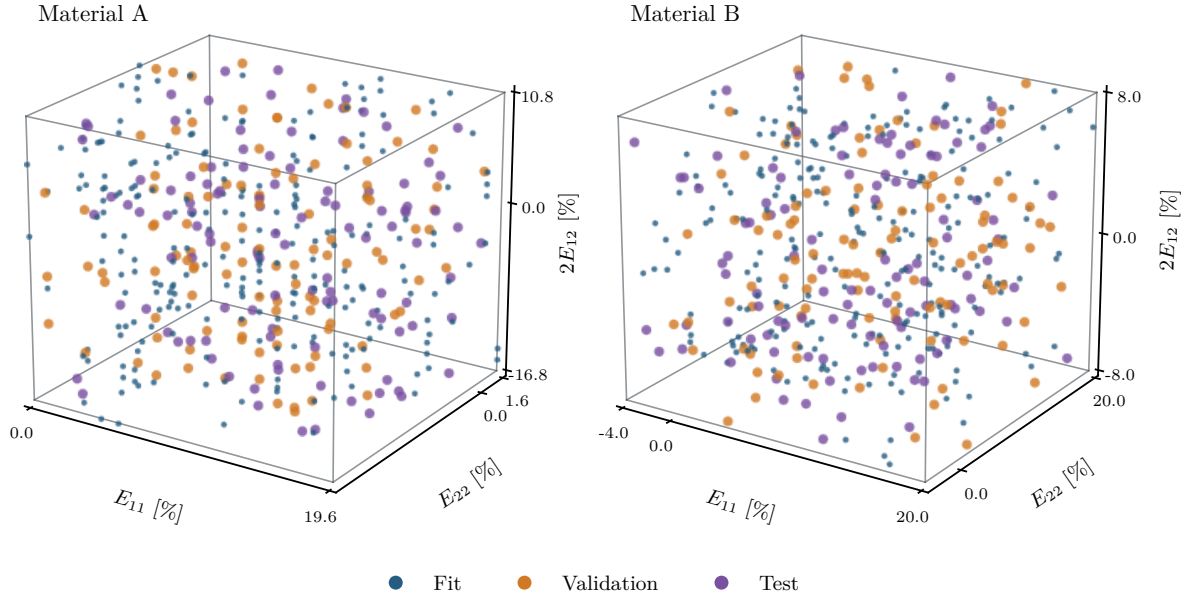


Figure 7: Strain-sampling domains for Material A (left) and Material B (right), in percent. Wireframes show the prescribed limits; colors identify fit, validation, and independent test. For legibility, each panel displays 240 fit, 110 validation, and 110 test coordinates selected deterministically; full counts appear in the text.

Optimization and model selection. For material A, each network represents one scalar energy. Its first strain derivative supplies stress, and its second strain derivative at the undeformed state supplies the reference tangent (initial stiffness); these are not independent outputs. We fit all three quantities against RVE data. Stress has the largest weight because it is the response queried throughout the FE^2 calculation. The loss combines dimensionless relative mean-square errors for stress, energy, and reference tangent, weighted 1, 0.2, and 0.02, respectively. The tangent target comes from a separate coarse periodic RVE calculation.

Parameter fitting starts with 2600 full-batch Adam steps and ends with up to 400 L-BFGS iterations to refine the fit. Validation stress error determines which checkpoint is retained; the independent test and probe states are not consulted. The selected ICNN and ICKAN checkpoints both use random-initialization seed 16 and 32 paired features. Each has two hidden layers, with 24 nodes per layer for ICNN and 12 for ICKAN. Because the earlier baselines used different features, objectives, and training budgets, their comparison measures practical accuracy under those protocols, not compute-matched architectural superiority.

Before entering either neural core, each feature z_i is centered on its undeformed value z_i^0 and divided by a positive scale s_i . The center is updated as the learned feature powers change; the scale $s_i = \max(10^{-4}, \max_{\text{fit}} |z_i - z_i^0|)$ is set once from the initial spread over fitting states. This keeps input magnitudes comparable during optimization without changing the constitutive constraints. For the selected ICKAN, each connection learns coefficients for six integrated-hat basis functions; their knot centers are fixed from -1.2 to 1.2 at spacing $h = 0.48$.

Material B supplies the controlled feature comparison absent from A. We trained Free and the fixed- and learned-feature versions of ICNN and ICKAN at seeds 16, 29, and 47. Within each core, fixed and learned variants share the 32-feature initialization, data, and objective. Checkpoints were selected by validation stress error and locked before test and loading-path evaluation. All runs used validation-plateau learning-rate scheduling and stopping, then L-BFGS, with a 200,000-step Adam cap. Free reached this cap in every seed without a plateau; its Adam stage is therefore budget-limited.

5.3 Constitutive response and predictive accuracy

For material A, the selected ICNN and ICKAN models attain held-out aggregate stress errors of approximately 0.65%; Table 3 reports the independently audited selected checkpoints. Free has a corresponding test error of 1.2643%. This comparison concerns the archived training protocols, not a matched-budget architecture study. Outside the training domain, Free has a lower aggregate probe error than either polyconvex model; Supplementary Section S1 preserves those results. Architectural admissibility does not guarantee extrapolation accuracy.

Table 3: Material A: independent-test constitutive errors of the selected polyconvex models, in percent.

Model	e_S^{test}	e_W^{test}	Worst test
ICNN	0.6491	0.2169	5.53
ICKAN	0.6551	0.2148	5.78

All values are percentages. Stress and energy errors are aggregate array norms; the last column is the maximum per-state stress error. Material A: 400 independent test states.

Draft task—evidence pending. Add FOM/model stress curves on predeclared axial, shear and combined loading paths, followed by the corresponding independent-test results for B. Inventory existing evaluations before requesting new FOM solves; do not connect unrelated scattered states as though they were a loading path. Report componentwise behavior and variability where needed to interpret aggregate errors.

5.4 Contribution of learned paired features

Draft task—evidence pending. Run a controlled comparison of fixed and learned admissible paired features, preferably for both A and B. Keep the core architecture, feature count, data, objective and optimization budget comparable; declare the fixed configuration and multiple initialization seeds before inspecting test errors. Retain model-specific reference normalization and growth terms. This comparison is not supplied by the archived baselines and has no results yet.

5.5 Mechanical consistency and scope

The following numerical checks concern the available material-A checkpoints. They complement, rather than replace, the construction guarantees in Section 4.

The archived independent audit records successful energy first- and second-derivative checks and saved-weight nonnegative-energy sufficient bounds for both selected models. The conservative finite-interval margins are 17.25 and 16.82 in normalized energy units. Random admissible-deformation tests support implementation consistency, but the polyconvexity argument is analytical. Neither aggregate error nor a successful bound on energy is a uniform bound on stress prediction error.

For the energy-based models, sampled rank-one curvatures provide an implementation check using the stress-dependent expression derived in Appendix A. The detailed sampling audit is retained in Supplementary Section S2. These finite samples do not establish polyconvexity; the ICNN and ICKAN guarantee follows from the analytical construction.

Closed-cycle work consistency follows from the scalar potential for Free, ICNN, and ICKAN. A numerical audit of the material-A constitutive outputs is reported in Supplementary Section S3.

Draft task—evidence pending. Repeat the reference-state, derivative, cycle and saved-parameter nonnegative-energy checks for the selected material-B models. Report bounds that fail or remain inconclusive. Existing A-checkpoint audits cannot certify B, and a second material trained separately does not demonstrate transfer between geometries.

6 Structural deployment for material A: accuracy and online cost

6.1 Purpose, boundary conditions, and coverage

We now use material A to compare learned constitutive laws and reduced microscopic solvers within the same nonlinear structural calculation. The coupon measures how their approximations affect structural accuracy and online cost. The dog-bone profile has overall length 165 mm, grip width 19 mm, narrow-section width 13 mm, narrow-section length 57 mm, and fillet radius 76 mm. It is a plane-strain computational benchmark using an ASTM-inspired in-plane profile, not a simulation certified to reproduce an ASTM tensile test.

The two end faces receive equal-and-opposite axial resultants of 100 kN per end, applied in 20 equal increments. Three point constraints remove rigid modes: the axial and transverse displacement at the left mid-edge node, and transverse displacement at the right mid-edge node. Entire end faces are not clamped. Consequently, a small skew of the deformed ends is not ruled out by the boundary conditions. The anisotropic response can couple axial extension to shear.

The macro mesh contains 690 quadratic triangles, 1501 nodes, and 2070 integration points. The reference extrusion thickness used to convert the two-dimensional stress integrals into the stated force resultants is 50 mm. This force convention is part of the plane-strain benchmark, not an ASTM specimen-thickness specification. The physical cell size is declared as 1.3 mm, corresponding to ten cells across the narrow width. This provides a scale-separation assumption, not a direct numerical validation of homogenization error at that ratio. All saved converged integration-point strains for every reported model remain inside the unchanged training box. This check does not automatically cover unsaved intermediate Newton iterates or certify another load case.

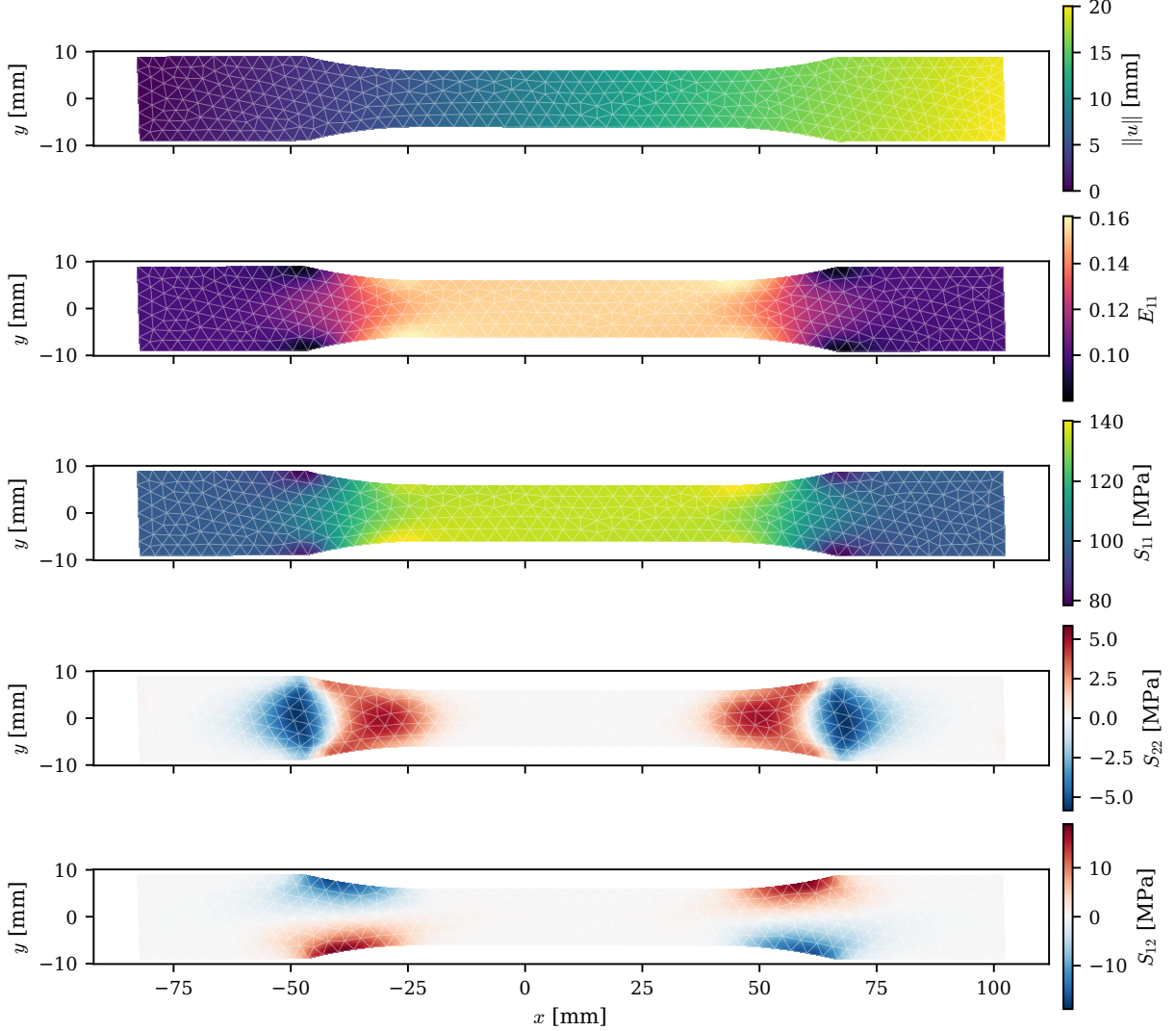


Figure 8: Nested FOM-FE² reference at 100 kN per end. Fields are displayed on the actual deformed mesh, without displacement amplification. Gauss-point strain and stress values are first averaged within each element and then patch-averaged to nodes for visualization only. These recovered contours are not the raw fields used in error norms and can smooth local extrema. Stresses are second Piola components. A three-dimensional von Mises stress is not reconstructed because the saved output does not include the necessary out-of-plane Cauchy stress.

6.2 Reduced-state representation for material A

The 4950 displacement snapshots provide the basis of the reduced microscopic models as well as the effective constitutive labels. Figure 9 separates POD compression from the strain-informed coordinate fit. The relative snapshot truncation norm for the retained 39-dimensional space is 9.94×10^{-7} . After its orthogonal rotation into primary and secondary spaces, $n = 3$ and $\bar{n} = 36$ in the notation of Section 3.

The fitted relation between the deployed primary coordinates and macroscopic strain has an aggregate

relative array discrepancy of 3.60×10^{-5} on these snapshots; its maximum absolute component discrepancy is 2.55×10^{-5} . These quantities explain why macroscopic strain is a useful primary-coordinate predictor for this material. They do not imply exact equality at unsampled states, nor do they measure the closure error or the equilibrium residual. In particular, the direct substitution in D-HPROM-ANN is an additional approximation, even when the plot appears visually indistinguishable from the identity.

The secondary-coordinate closure is a 3–64–64–36 feed-forward network with two tanh hidden layers and a linear output. Its training script uses input standardization, the raw secondary-coordinate loss in Eq. (23), and a 90/10 internal fit/validation split with seed 20260903. Full-batch Adam has initial learning rate 2×10^{-3} , a validation-driven reduction schedule, and a maximum budget of 20000 epochs. The archived best relative validation mean-square error is 3.75×10^{-7} . This is an internal closure metric after a POD built from all 4950 snapshots, not an independent material-generalization estimate. Independent effective-stress and structural tests provide the latter assessment.

Each deployed adaptive cubature field has architecture 3–128–128–128–10, GELU hidden activations, and the scaled softmax output of Eq. (39). The residual and stress checkpoints use distinct ten-element supports. Their larger neural cores illustrate why the online cost cannot be inferred solely from the three equilibrium coordinates or from the element count. These reduction-training choices differ from those of the constitutive networks; the common material and data domain do not imply matched offline training budgets.

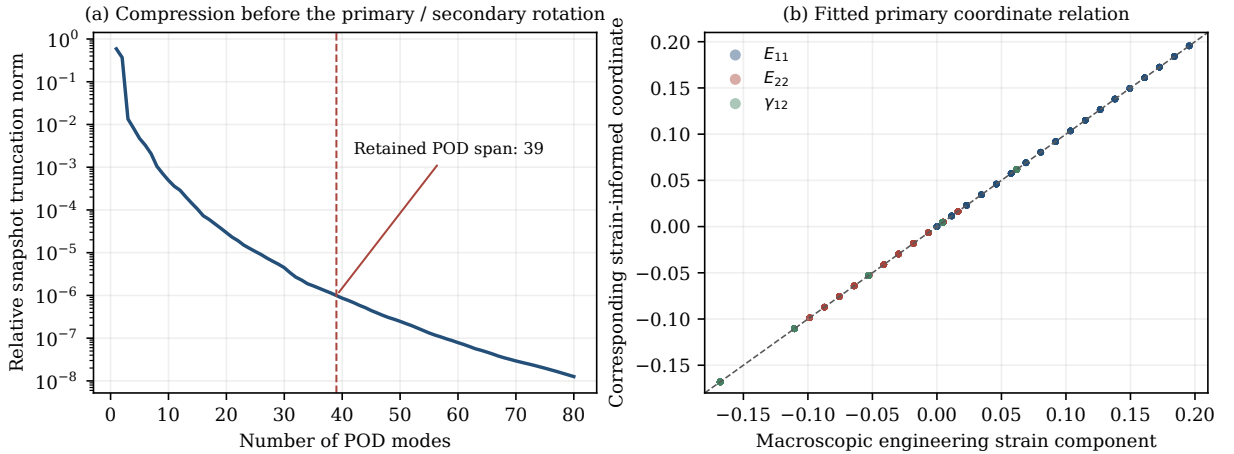


Figure 9: Diagnostics from the deployed 39-mode basis artifact. Left: relative POD snapshot truncation norm, before the primary/secondary rotation. Right: strain-informed primary coordinates versus the corresponding macroscopic strains; every tenth snapshot is plotted, whereas the reported discrepancies use all 4950 states. Neither diagnostic measures structural accuracy or proves validity outside the sampled domain.

6.3 Online models and implementation verification

The linear HPROM uses $n_{\text{tra}} = 39$ generalized coordinates, a residual rule supported on 135 elements, and a stress rule on 73 elements; their union contains 183 elements. The integrand-SVD tolerances are $\varepsilon_{\text{SVD}}^r = 10^{-3}$ and $\varepsilon_{\text{SVD}}^s = 10^{-5}$, retaining 134 and 72 integrand modes, respectively. The greedy integration tolerance in Eq. (36) is $\varepsilon_{\text{ECM}} = 10^{-6}$ for both rules, including the multiplier-sum constraint.

The iterative HPROM-ANN uses $n = 3$ primary and $\bar{n} = 36$ secondary coordinates, with ten residual and ten stress elements and a union of 19. Initial fixed-ECM candidates use $\varepsilon_{\text{SVD}} = 10^{-4}$ and $\varepsilon_{\text{ECM}} = 10^{-6}$; adaptive pruning targets ten elements per rule. Thus this final count is a prescribed pruning target assessed through integration and response errors, not a consequence of the greedy tolerance alone. The support-generation driver uses up to 500 sampled states and tests up to 20 removal candidates per elimination. Its positivity-enforcing phase is active from the first elimination; the available driver does not enable or supply the global graph coupling. The residual weight field is subsequently refitted on the same pruned support with a longer training budget. Historical pruning diagnostics are not stored in the deployed checkpoints, so their filenames alone are not evidence of a graph-regularized run.

The direct model evaluates stress on ten elements and solves no microscopic equilibrium equations. All these counts refer to reduced computational meshes, not full-mesh evaluations followed by masking. Figure 10 shows their spatial distribution and the overlap of residual and output rules.

For the optimized linear and direct implementations, the maximum normalized stress differences from the original implementations over the verification states are 4.71×10^{-14} and 4.20×10^{-13} . Corresponding

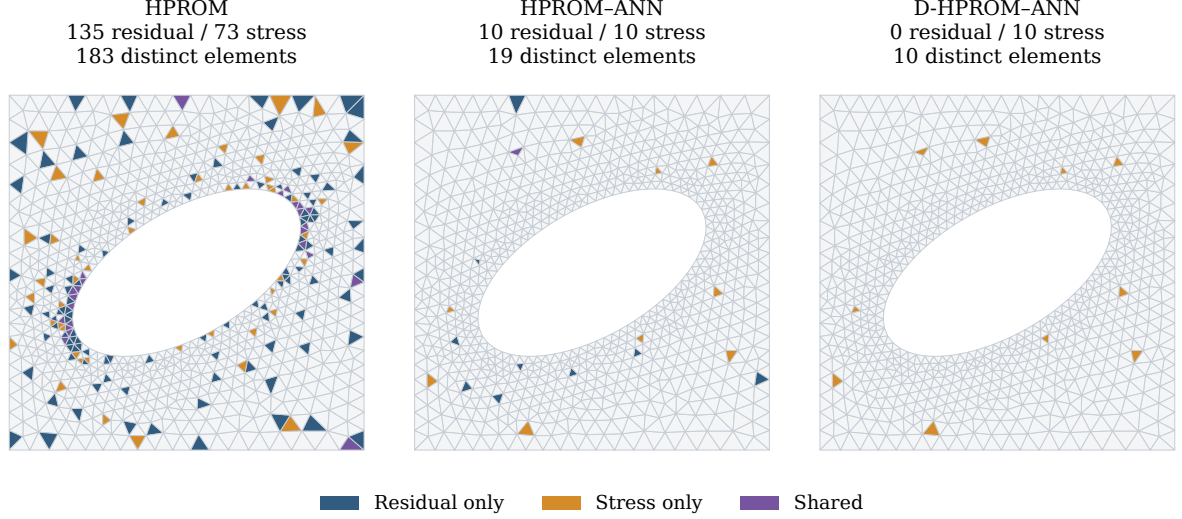


Figure 10: Element supports of the deployed hyperreduced microscopic models. The pale full-cell mesh provides location context only; discarded elements are not assembled online. Colors distinguish residual-only, stress-only, and shared elements. Supports are read from the same index arrays used to construct the reduced MDPA meshes.

tangent differences are 2.68×10^{-10} and 7.25×10^{-10} . Independent finite-difference checks give discrepancies of approximately 3.51×10^{-9} and 2.38×10^{-9} . Such comparisons check that optimization preserves the deployed model; they do not establish that either model exactly reproduces full microscopic equilibrium.

6.4 Error and timing definitions

For a saved nodal or integration-point array A , the reported relative error is

$$e_A = 100 \frac{\|\text{vec}(A - A_0)\|_2}{\|\text{vec}(A_0)\|_2}, \quad (80)$$

where A_0 is the nested reference on the same meshes. These are unweighted discrete array norms. The strain array stores engineering shear, whereas the stress array stores tensor shear. They are neither quadrature-weighted continuum L^2 norms nor invariant tensor Frobenius norms. The metric is kept identical across models; this does not make the three different field errors interchangeable.

Online wall time includes the complete macroscopic solve, microscopic or surrogate stress/tangent evaluations, communication, and lazy worker construction within the timed solve. Parent setup, offline data generation, training, reduced-mesh construction outside the timed region, and result export are excluded. The FOM uses 20 worker processes with one numerical-library thread per worker. The three optimized microscopic surrogates use the same process configuration and three sequential fresh complete runs. PANN evaluations are batched in one process and use the separately investigated thread settings shown in Table 4.

The macroscopic solve stops when the free-degree-of-freedom residual norm falls below 10^{-5} N or its ratio to the initial increment residual falls below 10^{-7} . The periodic FOM uses a relative microscopic displacement-increment tolerance of 10^{-10} , with continuation when needed. The archived FOM and selected PANN times each have one repetition in this comparison. Thus their ratios are observed online speedups under the stated implementations, not matched-thread efficiency estimates or repeated-run confidence intervals. Table 5 exposes the available reduced-model variability rather than selecting its fastest repetition. A publication-final timing study should repeat every tier under a common resource-budget protocol and retain all repetitions.

The affine HPROM uses its exact reduced residual Jacobian, whereas HPROM-ANN uses the projected-stiffness modified-Newton matrix described in Section 3.4.2; both use a microscopic increment tolerance of 10^{-10} . Their effective constitutive tangents nevertheless account for the equilibrium sensitivity of their respective full residuals. The archived structural runs do not retain microscopic iteration totals. The reported speedup therefore compares these implemented workflows, including their different iteration matrices and derivative costs; it does not isolate a nonlinear-representation gain or quantify the iteration penalty of modified Newton.

Table 4: Complete coupon simulations: final-field errors against the nested FOM and archived online wall times.

Model	CPU	n_{run}	t [s]	t_0/t	e_u [%]	e_E [%]	e_S [%]
FOM	20p	1	4268.679	1.0	0.00000	0.00000	0.00000
HPROM	20p	3	67.348	63.4	0.00565	0.01653	0.00099
HPROM-ANN	20p	3	30.143	141.6	0.05200	0.14968	0.03727
D-HPROM-ANN	20p	3	3.788	1126.9	0.02324	0.04851	0.00967
ICNN	4t	1	1.734	2461.3	0.42468	0.77028	0.08880
ICKAN	8t	1	3.669	1163.5	0.38657	0.78932	0.07800
Free	12t	1	1.356	3146.9	0.92988	1.49464	0.13492
Regression	16t	1	0.946	4510.4	0.60620	0.81218	0.07378

p: worker processes, one numerical-library thread each; t: PyTorch threads. For $n_{\text{run}} = 3$, time is the median; for $n_{\text{run}} = 1$, it is the archived single run. These are online wall times, not offline-inclusive or matched-thread speedups.

Table 5: Sequential optimized-run repetitions used for median HPROM timings.

Model	t_1 [s]	t_2 [s]	t_3 [s]
HPROM	63.354	67.348	68.953
HPROM-ANN	27.947	34.628	30.143
D-HPROM-ANN	3.788	3.758	3.969

6.5 Accuracy–cost trade-off and Newton behavior

Figure 11 and Table 4 show distinct operating points rather than a single universally best model. The linear HPROM gives the smallest displacement error, approximately 0.00565%, at a median online time of 67.35 s. The iterative HPROM-ANN reduces this time to 30.14 s with a displacement error of approximately 0.0520%. The optimized iterative nonlinear model is therefore about 2.23 times faster than the *optimized* linear model; comparison against the old linear implementation would overstate this gain. The direct closure reduces the time to 3.79 s and has a displacement error of approximately 0.0232% on this path. Its lower error than the iterative nonlinear model is an observed outcome, not a consequence of a general error ordering.

The ICNN and ICKAN times are 1.73 s and 3.67 s, with displacement errors of approximately 0.425% and 0.387%. Their finite-strain structural accuracy is therefore consistent with a useful constitutive approximation, although it is poorer than the reduced microscopic models on this benchmark. Direct stress regression is fastest, and Free also converges. These baselines establish the accuracy and cost achievable on this loading path alongside the constrained models.

All models require four macroscopic iterations per increment in the archived runs. Consequently, this example provides no empirical evidence that polyconvexity reduced Newton iterations relative to Free or Regression. It establishes that the selected constrained models can be used successfully with the common structural solver, while maintaining their constitutive guarantees. A separate challenge problem would be needed to establish a solver-robustness advantage, and its reference solution would need to be validated independently.

7 Conclusions

Paired directional features with constrained trainable powers and determinant exponents provide a practical polyconvex constitutive construction for the anisotropic periodic material studied here. Their isotropic reference derivatives allow exact stress normalization through an affine determinant correction without tying the normalization coefficient to the volumetric barrier stiffness. The proof combines feature convexity in independent minors with core convexity and monotonicity, and includes every correction in the complete energy. Global nonnegative energy is supported separately by a sufficient bound evaluated for the selected parameters. Both a smooth ICNN and an integrated-hat spline-ICKAN achieve approximately 0.65% held-out aggregate stress error and below 0.43% displacement error in the nominal coupon simulation.

Optimized reduced microscopic models provide a complementary accuracy–cost reference: they are more accurate on this coupon, while direct learned constitutive evaluation is generally faster. All compared models converge with the same number of macroscopic iterations per increment. The structural results therefore demonstrate the practical use of the constrained energies alongside their analytical guarantees; they establish neither a Newton-convergence advantage nor a universal accuracy ranking.

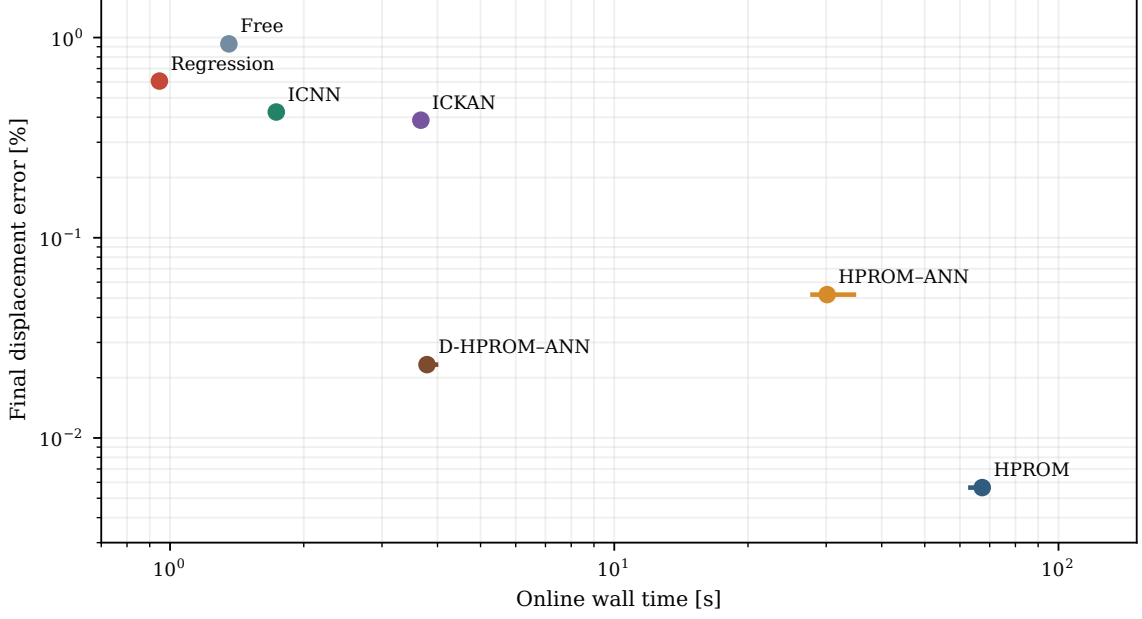


Figure 11: Online cost versus final displacement error on the common coupon problem. Horizontal segments show the observed range of three optimized HPROM runs; unbarred points are archived single PANN runs. Process and thread counts differ as detailed in Table 4. The plot does not include offline costs.

A Rank-one paths and finite-strain energy curvature

This appendix supplies the algebra behind the two finite-strain curvature statements in criterion (C5). For a twice differentiable energy, nonnegative curvature along every rank-one perturbation is expressed by the Legendre–Hadamard condition

$$\partial_{\mathbf{F}\mathbf{F}}^2 W(\mathbf{F})[\mathbf{a} \otimes \mathbf{b}, \mathbf{a} \otimes \mathbf{b}] \geq 0 \quad \text{for all } \mathbf{a}, \mathbf{b} \in \mathbb{R}^2. \quad (81)$$

The argument below shows why the polyconvex representation in Eq. (52) implies this condition.

Determinant along a rank-one path. For arbitrary $\mathbf{F}, \mathbf{H} \in \mathbb{R}^{2 \times 2}$, expansion of the two products in the determinant gives

$$\begin{aligned} \det(\mathbf{F} + t\mathbf{H}) &= (F_{11} + tH_{11})(F_{22} + tH_{22}) - (F_{12} + tH_{12})(F_{21} + tH_{21}) \\ &= \det \mathbf{F} + t(F_{22}H_{11} + F_{11}H_{22} - F_{21}H_{12} - F_{12}H_{21}) + t^2 \det \mathbf{H}. \end{aligned} \quad (82)$$

The coefficient of t is $\text{cof } \mathbf{F} : \mathbf{H}$, since

$$\text{cof } \mathbf{F} = \begin{pmatrix} F_{22} & -F_{21} \\ -F_{12} & F_{11} \end{pmatrix}, \quad \mathbf{A} : \mathbf{B} = \sum_{i,j} A_{ij} B_{ij}. \quad (83)$$

For $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, its entries are $H_{ij} = a_i b_j$, so

$$\det \mathbf{H} = a_1 b_1 a_2 b_2 - a_1 b_2 a_2 b_1 = a_1 a_2 (b_1 b_2 - b_2 b_1) = 0. \quad (84)$$

The determinant is therefore affine along a rank-one path:

$$\det(\mathbf{F} + t\mathbf{a} \otimes \mathbf{b}) = \det \mathbf{F} + t \text{cof } \mathbf{F} : (\mathbf{a} \otimes \mathbf{b}). \quad (85)$$

If $\mathbf{F}_B = \mathbf{F}_A + \mathbf{H}$ and $\mathbf{F}_s = (1-s)\mathbf{F}_A + s\mathbf{F}_B$, with $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, then

$$(\mathbf{F}_s, \det \mathbf{F}_s) = (1-s)(\mathbf{F}_A, \det \mathbf{F}_A) + s(\mathbf{F}_B, \det \mathbf{F}_B). \quad (86)$$

For $0 \leq s \leq 1$ and positive endpoint determinants, the whole segment remains admissible. Convexity of $\mathcal{P}$ therefore yields

$$W(\mathbf{F}_s) \leq (1-s)W(\mathbf{F}_A) + sW(\mathbf{F}_B). \quad (87)$$

For a twice differentiable energy, the restriction to this line consequently has nonnegative second derivative, as required by Eq. (81).

Strain curvature and the chain rule. Let $\mathbf{H} \in \mathbb{R}^{2 \times 2}$ be an arbitrary perturbation and let t parametrize $\mathbf{F}(t) = \mathbf{F} + t\mathbf{H}$. Dots denote derivatives with respect to t , and $\text{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$ denotes the symmetric part of a matrix. Direct multiplication gives

$$\begin{aligned} \mathbf{E}(t) &= \frac{1}{2} [\mathbf{F}^T \mathbf{F} - \mathbf{I} + t(\mathbf{F}^T \mathbf{H} + \mathbf{H}^T \mathbf{F}) + t^2 \mathbf{H}^T \mathbf{H}] \\ &= \mathbf{E}(0) + t \text{sym}(\mathbf{F}^T \mathbf{H}) + \frac{1}{2} t^2 \mathbf{H}^T \mathbf{H}. \end{aligned} \quad (88)$$

Thus $\dot{\mathbf{E}}(0) = \text{sym}(\mathbf{F}^T \mathbf{H})$ and $\ddot{\mathbf{E}}(0) = \mathbf{H}^T \mathbf{H}$. Using the engineering-strain convention of Eq. (2), the chain and product rules give

$$\begin{aligned} \frac{dW}{dt} &= \mathbf{s} \cdot \dot{\mathbf{e}}, \\ \frac{d^2W}{dt^2} &= \sum_{i,j} \frac{\partial s_i}{\partial e_j} \dot{e}_j \dot{e}_i + \sum_i s_i \ddot{e}_i = \dot{\mathbf{e}}^T \mathbf{D} \dot{\mathbf{e}} + \mathbf{S} : \ddot{\mathbf{E}}. \end{aligned} \quad (89)$$

Substituting $\ddot{\mathbf{E}}(0)$ gives

$$\left. \frac{d^2W(\mathbf{F} + t\mathbf{H})}{dt^2} \right|_{t=0} = \dot{\mathbf{e}}(0)^T \mathbf{D} \dot{\mathbf{e}}(0) + \mathbf{S} : (\mathbf{H}^T \mathbf{H}). \quad (90)$$

Here $\mathbf{D} = \partial \mathbf{s} / \partial \mathbf{e}$ is the strain tangent and $\mathbf{S}$ is the second Piola–Kirchhoff stress, both evaluated at $t = 0$. The first term accounts for the change of stress with strain; the second accounts for the curvature of the strain path, weighted by the current stress. Even if $\mathbf{D}$ is positive semidefinite, the stress term can be negative in compression.

An analytical counterexample. Consider $W(\mathbf{E}) = \mu \mathbf{E} : \mathbf{E}$, where $\mu > 0$ is a constant with units of energy density. One has $\mathbf{S} = 2\mu \mathbf{E}$. Since

$$W(\mathbf{e}) = \mu(e_1^2 + e_2^2 + e_3^2/2), \quad \mathbf{D} = \text{diag}(2\mu, 2\mu, \mu), \quad (91)$$

the energy is strictly convex in engineering strain. At $\mathbf{F} = \lambda \mathbf{I}$, where $\lambda > 0$ is the common in-plane stretch, choose the rank-one perturbation $\mathbf{H} = \text{diag}(1, 0)$. Then

$$\dot{\mathbf{e}} = (\lambda, 0, 0)^T, \quad \mathbf{S} = \mu(\lambda^2 - 1)\mathbf{I}, \quad \mathbf{H}^T \mathbf{H} = \text{diag}(1, 0). \quad (92)$$

Hence the strain-tangent term is $2\mu\lambda^2$, and the stress term is $\mu(\lambda^2 - 1)$, giving

$$\left. \frac{d^2W(\lambda \mathbf{I} + t\mathbf{H})}{dt^2} \right|_{t=0} = 2\mu\lambda^2 + \mu(\lambda^2 - 1) = \mu(3\lambda^2 - 1). \quad (93)$$

At $\lambda = 1/2$, the curvature is $-\mu/4$, violating Eq. (81) despite convexity in $\mathbf{E}$. This is an analytical counterexample, not a sampled observation about the RVE or a trained model. Compression alone does not force negative curvature: at $\lambda = 3/4$, the same expression is $11\mu/16 > 0$.

Equivalently, direct substitution gives

$$W(\lambda \mathbf{I} + t\mathbf{H}) = \frac{\mu}{4} [(\lambda + t)^2 - 1]^2 + \frac{\mu}{4} (\lambda^2 - 1)^2, \quad (94)$$

whose second derivative at zero is $\mu(3\lambda^2 - 1)$. The path is admissible near zero because $\det(\lambda \mathbf{I} + t\mathbf{H}) = \lambda(\lambda + t) > 0$ for $t > -\lambda$.

B Proofs for the paired features and complete energy

The proof follows the construction used in the main text. We first isolate one stretch–determinant contribution and derive the exponent bounds that make it convex. We then assemble two perpendicular contributions and compute their reference derivative, proving Proposition 1. Finally, we compose the resulting features with the monotone convex core and examine W_{learn} , W_{corr} , W_{vol} , and W_{dist} to establish Proposition 2.

B.1 Joint convexity of the directional building block

Every contribution in Eq. (59) has the form

$$f(\mathbf{x}, J) = \|\mathbf{x}\|^a J^{-b}, \quad \mathbf{x} = \mathbf{F}\mathbf{d}, \quad (95)$$

up to a positive constant, with $a = 2p$ or $2q$. It is therefore enough to determine when this building block is jointly convex in $(\mathbf{x}, J)$. Let $t = \|\mathbf{x}\| > 0$. Because f depends on $\mathbf{x}$ only through t , its Hessian separates into a block coupling the radial variable t to J and transverse directions orthogonal to $\mathbf{x}$. The radial-determinant block is

$$\begin{pmatrix} a(a-1)t^{a-2}J^{-b} & -abt^{a-1}J^{-b-1} \\ -abt^{a-1}J^{-b-1} & b(b+1)t^aJ^{-b-2} \end{pmatrix}. \quad (96)$$

For this symmetric 2×2 block to be positive semidefinite, its diagonal entries and determinant must be nonnegative. The diagonal entries are nonnegative for $a \geq 1$ and $b \geq 0$, while the determinant is

$$\begin{aligned} \det &= ab[(a-1)(b+1) - ab]t^{2a-2}J^{-2b-2} \\ &= ab(a-b-1)t^{2a-2}J^{-2b-2} \geq 0. \end{aligned} \quad (97)$$

The factor $a-b-1$ gives the upper bound $b \leq a-1$; if this bound is violated, the determinant is negative and the block has curvatures of opposite sign. The remaining, transverse Hessian eigenvalues are $at^{a-2}J^{-b} \geq 0$. Hence $a \geq 1$ and $0 \leq b \leq a-1$ make the complete Hessian positive semidefinite. Continuous extension gives convexity at $\mathbf{x} = 0$. For the constitutive evaluation on $\text{GL}^+(2)$, a nonzero reference direction cannot satisfy $\mathbf{F}\mathbf{d} = 0$, so this point does not introduce an interior singularity of the directional norm. Substituting $a = 2p$ gives $p \geq \frac{1}{2}$ and $0 \leq b \leq 2p-1$; the same argument with (q, c) establishes all bounds in Eq. (60).

B.2 Proof of Proposition 1

Proof. Convexity and objectivity. For any fixed learned parameter set, the directions $\mathbf{d}_i$ and $\mathbf{d}_i^\perp$ are independent of deformation. The maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$ are therefore linear. Composing the convex building block above with these maps, multiplying by the positive factors $1/p_i$ and $1/q_i$, and adding the two contributions proves convexity of z_i in independent $(\mathbf{F}, J)$. A left proper rotation changes $\mathbf{F}\mathbf{d}$ to $\mathbf{Q}\mathbf{F}\mathbf{d}$ without changing its norm, so each contribution is objective.

Reference value and derivative. We now return to the physical relation $J = \det \mathbf{F}$, still holding every learned parameter fixed. At $\mathbf{E} = \mathbf{0}$, both directional invariants and J equal one, so the two contributions give $z_i^0 = 1/p_i + 1/q_i$. Moreover, $I_4(\mathbf{d}_i) = \mathbf{d}_i^T(\mathbf{I} + 2\mathbf{E})\mathbf{d}_i$ and $J^2 = (1 + 2E_{11})(1 + 2E_{22}) - 4E_{12}^2$ imply $\partial_{\mathbf{E}} I_4(\mathbf{d}_i) = 2\mathbf{d}_i \otimes \mathbf{d}_i$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$. Applying the product rule to the first contribution gives

$$\partial_{\mathbf{E}} \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} \Big|_0 = \frac{1}{p_i} [2p_i \mathbf{d}_i \otimes \mathbf{d}_i - b_i \mathbf{I}] = 2\mathbf{d}_i \otimes \mathbf{d}_i - \frac{b_i}{p_i} \mathbf{I}. \quad (98)$$

The prefactor $1/p_i$ has cancelled the factor p_i produced by differentiating the stretch power. The perpendicular contribution has the same form with $(\mathbf{d}_i, p_i, b_i)$ replaced by $(\mathbf{d}_i^\perp, q_i, c_i)$. The two directional projectors satisfy $\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp = \mathbf{I}$; adding the contributions therefore removes the directional dependence at the reference and gives

$$\partial_{\mathbf{E}} z_i|_0 = 2(\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp) - \left(\frac{b_i}{p_i} + \frac{c_i}{q_i} \right) \mathbf{I} = \rho_i \mathbf{I}. \quad (99)$$

In the work-conjugate engineering-strain convention of Eq. (2), this tensor gradient corresponds to $\partial_{\mathbf{e}} z_i|_0 = \rho_i(1, 1, 0)^T$, proving Eq. (62). $\square$

Positive affine conditioning. The raw feature is converted to the core input by $y_i = (z_i - z_i^{\text{cen}})/s_i$, with $s_i > 0$ independent of deformation. Writing $\mathbf{X} = (\mathbf{F}, J)$,

$$\partial_{\mathbf{X}\mathbf{X}}^2 y_i = \frac{1}{s_i} \partial_{\mathbf{X}\mathbf{X}}^2 z_i, \quad y_i(\mathbf{X}_B) - y_i(\mathbf{X}_A) = \frac{z_i(\mathbf{X}_B) - z_i(\mathbf{X}_A)}{s_i}. \quad (100)$$

The first identity preserves positive-semidefinite curvature and the second preserves componentwise order. Thus the numerical conditioning applied before the monotone core changes neither property used in the composition proof below.

B.3 Proof of Proposition 2

Proof. Potential structure, objectivity, and stress symmetry (C1)–(C3). Every directional norm and J is unchanged by left proper rotations. The features can also be expressed through $\mathbf{d}_i^T \mathbf{C} \mathbf{d}_i$ and $\sqrt{\det \mathbf{C}}$. Hence the complete energy is objective, and differentiation with respect to symmetric strain gives a symmetric material stress. Differentiating this scalar energy supplies (C1) and the major symmetry of its strain tangent. Smoothness holds on $\text{GL}^+(2)$ because $J > 0$, nonzero material directions have nonzero deformed length, and both cores are C^2 . Because this argument uses left rotations only, it does not impose a material symmetry group.

Reference normalization (C4). Equation (72) shows that W_{learn} sets the reference energy, and Eq. (74) shows that W_{corr} cancels the learned reference stress. The calculations for W_{vol} and W_{dist} in Section 4.5 show that both growth terms have zero value and zero strain gradient at the reference. Thus the complete energy satisfies both reference conditions.

Polyconvexity (C5). Let $\mathbf{G} \in \mathbb{R}^{2 \times 2}$ and $j > 0$ be independent. Define $\mathcal{P}(\mathbf{G}, j)$ by the right-hand side of Eq. (71), replacing $\mathbf{z}$ by $\mathbf{z}(\mathbf{G}, j)$, J by j , and $\text{tr } \mathbf{C}$ by $\|\mathbf{G}\|_F^2$. For two such pairs $\mathbf{X}_1 = (\mathbf{G}_1, j_1)$, $\mathbf{X}_2 = (\mathbf{G}_2, j_2)$ and $0 \leq t \leq 1$,

$$\begin{aligned} H(\mathbf{z}(t\mathbf{X}_1 + (1-t)\mathbf{X}_2)) &\leq H(t\mathbf{z}(\mathbf{X}_1) + (1-t)\mathbf{z}(\mathbf{X}_2)) \\ &\leq tH(\mathbf{z}(\mathbf{X}_1)) + (1-t)H(\mathbf{z}(\mathbf{X}_2)). \end{aligned} \quad (101)$$

The first inequality uses feature convexity and core monotonicity; the second uses core convexity. Therefore W_{learn} is convex up to its constant reference subtraction. The correction $W_{\text{corr}} = -r(j-1)$ is affine for either sign of r , while W_{vol} and W_{dist} are convex in independent $(\mathbf{G}, j)$ because their nonaffine parts are positive multiples of $-\log j$, $(j-1)^2$, and $\|\mathbf{G}\|_F^2$. Thus $\mathcal{P}$ is convex and $W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F})$, establishing Eq. (52).

Growth (C6). Convexity of H gives its supporting-plane inequality at $\mathbf{z}^0$:

$$H(\mathbf{z}) \geq H(\mathbf{z}^0) + \sum_i h_i(z_i - z_i^0) \geq H(\mathbf{z}^0) - \sum_i h_i z_i^0. \quad (102)$$

The second inequality uses $h_i, z_i \geq 0$. Hence W_{learn} has a deformation-independent lower bound and cannot cancel the analytic growth terms. As $J \rightarrow 0^+$, the logarithmic part of $W_{\text{vol}} + W_{\text{dist}}$ diverges while W_{corr} remains bounded. As $J \rightarrow \infty$, the positive quadratic part of W_{vol} dominates every potentially negative linear or logarithmic term. Finally, if J remains in a compact subset of $(0, \infty)$ while $\|\mathbf{F}\|_F \rightarrow \infty$, the term $\varepsilon \|\mathbf{F}\|_F^2/2$ in W_{dist} diverges and all determinant-only terms remain bounded. These are the three stated growth limits. $\square$

C Sufficient lower bound for nonnegative energy

Define $k_i = 1/p_i + 1/q_i$ and $\eta_i = \rho_i/k_i$. Write $t = \|\mathbf{F} \mathbf{d}_i\|^2$, $u = \|\mathbf{F} \mathbf{d}_i^\perp\|^2$. Hadamard's determinant inequality gives $tu \geq J^2$. Since each feature is increasing in t, u , its minimum at fixed J occurs on $tu = J^2$. Minimizing

$$\frac{t^{p_i}}{p_i J^{b_i}} + \frac{J^{2q_i - c_i} t^{-q_i}}{q_i} \quad (103)$$

gives $t^{p_i+q_i} = J^{2q_i+b_i-c_i}$ and hence $z_i \geq k_i J^{\eta_i}$. This is a separate lower bound for each feature; simultaneous attainability for all directions is not assumed. It therefore remains valid after summing the supporting-plane inequalities.

Together with $\text{tr } \mathbf{C}/2 \geq J$ and the supporting plane of H , this feature bound gives

$$W \geq g(J) := \sum_i h_i k_i [J^{\eta_i} - 1 - \eta_i(J-1)] + (\alpha + \varepsilon)(J-1 - \log J) + \frac{\beta}{2}(J-1)^2. \quad (104)$$

Because $g(1) = g'(1) = 0$, it remains to establish $g''(J) \geq 0$ for every $J > 0$. Differentiation gives

$$J^2 g''(J) = \alpha + \varepsilon + \beta J^2 + \sum_i c_i^* J^{\eta_i}, \quad c_i^* = h_i k_i \eta_i (\eta_i - 1). \quad (105)$$

Only terms with $0 < \eta_i < 1$ can have negative coefficients. If $c_- = \sum_{c_i^* < 0} |c_i^*|$, the two inequalities

$$\alpha + \varepsilon + \sum_{\eta_i < 0} c_i^* - c_- \geq 0, \quad (106)$$

$$\beta + \sum_{\eta_i \geq 1} c_i^* - c_- \geq 0 \quad (107)$$

are sufficient on $0 < J \leq 1$ and $J \geq 1$, respectively. For the first region, $J^{\eta_i} \leq 1$ for negative-coefficient terms and $J^{\eta_i} \geq 1$ when $\eta_i < 0$. For the second region, divide $J^2 g''$ by J : all negative-coefficient powers then satisfy $J^{\eta_i-1} \leq 1$, while $J^{\eta_i-1} \geq 1$ for $\eta_i \geq 1$ and $\beta J \geq \beta$. Dropping the remaining nonnegative contributions gives the stated bounds. Their failure is inconclusive.

The selected parameters require a sharper check. The audit partitions $[10^{-4}, 10^4]$ into 4000 logarithmically spaced intervals. On each interval $[J_L, J_R]$, a positive power term is bounded below by the smaller endpoint value, whereas a negative-coefficient term is bounded below by $c_i^* J_R^{\eta_i}$, since its exponent lies in $(0, 1)$. The constant and quadratic terms are bounded below by $\alpha + \varepsilon$ and βJ_L^2 . Below the finite interval, the nonnegative negative-exponent terms increase and the magnitudes of the negative-coefficient terms decrease toward $J = 0$. Above it, factoring out J^{η_-} with $\eta_- = \max_{c_i^* < 0} \eta_i < 1$ yields a lower bound whose retained positive powers increase. These endpoint and tail inequalities cover every $J > 0$, not only the grid points. Positive margins establish the sufficient analytical argument for the saved parameters to the accuracy of its floating-point evaluation; random energy samples alone would not establish it.

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